

4

Linear models

CONTENTS

4.1	Analysis of normal means	120
4.1.1	One-sample/paired analysis	120
4.1.2	Comparison of two normal means	121
4.2	Linear regression	124
4.2.1	Jeffreys prior	125
4.2.2	Gaussian prior	126
4.2.3	Continuous shrinkage priors	128
4.2.4	Predictions	129
4.2.5	Example: Factors that affect a home's microbiome	130
4.3	Generalized linear models	133
4.3.1	Binary data	135
4.3.2	Count data	137
4.3.3	Example: Logistic regression for NBA clutch free throws	138
4.3.4	Example: Beta regression for microbiome data	140
4.4	Random effects	141
4.5	Flexible linear models	149
4.5.1	Nonparametric regression	149
4.5.2	Heteroskedastic models	152
4.5.3	Non-Gaussian error models	153
4.5.4	Linear models with correlated data	153
4.6	Exercises	158

Linear models form the foundation for much of statistical modeling and intuition. In this chapter, we introduce many common statistical models and implement them in the Bayesian framework. We focus primarily on Bayesian aspects of these analyses including selecting priors, computation and comparisons with classical methods. The chapter begins with analyses of the mean of a normal population (Section 4.1.1) and comparison of the means of two normal populations (Section 4.1.2), which are analogous to the classic one-sample and two-sample t-tests. Section 4.2 introduces the more general Bayesian multiple linear regression model including priors that are appropriate for high-dimensional problems. Multiple regression is extended to non-Gaussian data in Section 4.3 via generalized linear models and correlated data in Section 4.4 via linear mixed models.

The Bayesian linear regression model makes strong assumptions including that the mean is a linear combination of the covariates, and that the observations are Gaussian and independent. Estimates are robust to small departures from these assumptions, but nonetheless these assumptions should be carefully evaluated (Section 5.6). Section 4.5 concludes with several extensions to the basic linear regression model to illustrate the flexibility of Bayesian modeling.

4.1 Analysis of normal means

4.1.1 One-sample/paired analysis

In a one-sample study, the n observations are modeled as $Y_i|\mu, \sigma^2 \stackrel{iid}{\sim} \text{Normal}(\mu, \sigma^2)$ and the objective is to determine if $\mu = 0$. This model is often applied to experiments where each unit i actually has two measurements taken under different conditions and Y_i is the difference between the measurements. For example, say student i 's math scores before and after a tutorial session are Z_{0i} and Z_{1i} , respectively, then testing if the (population) mean of $Y_i = Z_{1i} - Z_{0i}$ is zero is a way to evaluate the effectiveness of the tutorial sessions.

A Bayesian analysis specifies priors for μ and σ^2 and then summarizes the marginal posterior of μ accounting for uncertainty in σ^2 . In this chapter we use the Jeffreys' prior, but conjugate normal/inverse gamma priors can also be used as in Section 3.2. In most cases it is sufficient to plot the posterior density $p(\mu|\mathbf{Y})$ and report the posterior mean and 95% interval. We will also address the problem using a formal hypothesis test. In this chapter we will compute the posterior probabilities of the one-sided hypotheses $H_0 : \mu \leq 0$ and $H_1 : \mu > 0$; in Chapter 5 we test the point hypothesis $H_0 : \mu = 0$ versus $H_1 : \mu \neq 0$.

Conditional distribution with the variance fixed: Conditional on σ^2 , μ has Jeffreys prior $\pi(\mu) \propto 1$. The posterior is $\mu|\mathbf{Y} \sim \text{Normal}\left(\bar{Y}, \frac{\sigma^2}{n}\right)$ and thus the $100(1 - \alpha)\%$ posterior credible interval for μ is

$$\bar{Y} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad (4.1)$$

where z_τ is the τ quantile of the standard normal distribution (i.e., the normal distribution with mean zero and variance one). This exactly matches the classic $100(1 - \alpha)\%$ confidence interval. While the credible interval and confidence interval are numerically identical in this case, they are interpreted differently. The Bayesian credible interval quantifies uncertainty about μ given this dataset, $\mathbf{Y}$, whereas the confidence interval quantifies the anticipated variation in $\bar{Y}$ if we were to repeat the experiment.

For the test of null hypothesis $H_0 : \mu \leq 0$ versus the alternative hypothesis

$H_1 : \mu > 0$, the posterior probability of the null hypothesis is

$$\text{Prob}(H_0|\mathbf{Y}) = \text{Prob}(\mu < 0|\mathbf{Y}) = \Phi(-Z) \quad (4.2)$$

where Φ is the standard normal cumulative distribution function and $Z = \sqrt{n}\bar{Y}/\sigma$ and thus matches exactly with a frequentist p-value. By definition, $\Phi(z_\tau) = \tau$, and so the decision rule to reject H_0 in favor of H_1 if the posterior probability of H_0 is less than α is equivalent to rejecting H_0 if $-Z < z_\alpha$, or equivalently if $Z > z_{1-\alpha}$ ($-z_\alpha = z_{1-\alpha}$ due to the symmetry of the standard normal PDF). Therefore, the decision rule to reject H_0 in favor of H_1 at significance level α if $Z > z_{1-\alpha}$ is identical to the classic one-sided z-test. However, unlike the classical test, we can quantify our uncertainty using the posterior probability that the hypothesis H_0 (or H_1) is true since we have computed $\text{Prob}(H_0|\mathbf{Y})$.

Unknown variance: As shown in Section 2.3, the Jeffreys' prior for (μ, σ^2) is

$$\pi(\mu, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^{3/2}. \quad (4.3)$$

Appendix A.3 shows that the marginal posterior of μ integrating over σ^2 is

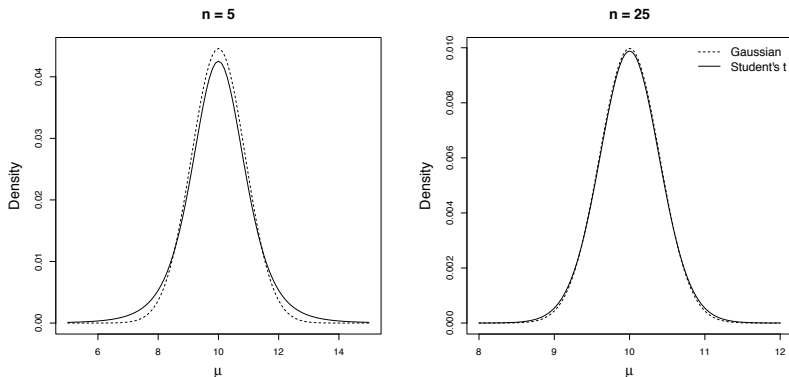
$$\mu|\mathbf{Y} \sim t_n(\bar{Y}, \hat{\sigma}^2/n), \quad (4.4)$$

where $\hat{\sigma}^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2/n$, i.e., the posterior is Student's t distribution with location $\bar{Y}$, variance parameter $\hat{\sigma}^2/n$ and n degrees of freedom. Posterior inference such as credible sets or the posterior probability that μ is positive follow from the quantiles of Student's t distribution. The credible set is slightly different than the frequentist confidence interval because the degrees of freedom in the classic t-test is $n - 1$, whereas the degrees of freedom in the posterior is n ; this is the effect of the prior on σ^2 .

In classical statistics, when σ^2 is unknown the Z-test based on the normal distribution is replaced with the t-test based on Student's t distribution. Similarly, the posterior distribution of the mean changes from a normal distribution given σ^2 to Student's t distribution when uncertainty about the variance is considered. Figure 4.1 compares the Gaussian and t density functions. The density functions are virtually identical for $n = 25$, but for $n = 5$ the t distribution has heavier tails than the Gaussian distribution; this is the effect of accounting for uncertainty in σ^2 .

4.1.2 Comparison of two normal means

The two-sample test compares the mean in two groups. For example, an experiment might take the blood pressure of a random sample of n_1 mail carriers that walk their route and n_2 mail carriers that drive their route to determine if these groups have different mean blood pressure. Let $Y_i \stackrel{iid}{\sim} \text{Normal}(\mu, \sigma^2)$ for $i = 1, \dots, n_1$ and $Y_i \stackrel{iid}{\sim} \text{Normal}(\mu + \delta, \sigma^2)$ for $i = n_1 + 1, \dots, n_1 + n_2 = n$, so that

**FIGURE 4.1**

Comparison of the Gaussian and Student's t distributions. Below are the Gaussian PDF with mean $\bar{Y}$ and standard deviation $\sigma/\sqrt{n}$ compared to the PDF of Student's t distribution with location $\bar{Y}$, scale $\hat{\sigma}/\sqrt{n}$, and n degrees of freedom. The plots assume $\bar{Y} = 10$, $\sigma = \hat{\sigma} = 2$, and $n \in \{5, 25\}$.

δ is the difference in means and the parameter of interest. Denote the sample mean of the n_j observations in group $j = 1, 2$ as $\bar{Y}_j$ and the group-specific variance estimators as $s_1^2 = \sum_{i=1}^{n_1} (Y_i - \bar{Y}_1)^2 / n_1$ and $s_2^2 = \sum_{i=n_1+1}^{n_1+n_2} (Y_i - \bar{Y}_2)^2 / n_2$.

Conditional distribution with the variance fixed: Conditional on the variance and flat prior $\pi(\mu, \delta) \propto 1$ it can be shown that the posterior of the difference in means is

$$\delta | \mathbf{Y} \sim \text{Normal} \left(\bar{Y}_2 - \bar{Y}_1, \sigma^2 \left[\frac{1}{n_1} + \frac{1}{n_2} \right] \right). \quad (4.5)$$

As in the one-sample case, posterior intervals and probabilities of hypotheses can be computed using the quantiles of the normal distribution. Also, as in the one-sample case, the credible set and rejection rule match the classic confidence interval and one-sided z -test numerically but have a different interpretations.

Unknown variance: The Jeffreys' prior for (μ, δ, σ^2) is (Section 2.3)

$$\pi(\mu, \delta, \sigma^2) \propto \frac{1}{(\sigma^2)^2}. \quad (4.6)$$

Appendix A.3 shows (as a special case of multiple linear regression, see Section 4.2) the marginal posterior distribution of δ integrating over both μ and σ^2 is

$$\delta | \mathbf{Y} \sim t_n \left[\bar{Y}_2 - \bar{Y}_1, \hat{\sigma}^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \right], \quad (4.7)$$

Listing 4.1

R code for comparing two normal means with the Jeffreys' prior.

```

1  # Y1 is the n1-vector of data for group 1
2  # Y2 is the n2-vector of data for group 2
3
4  # Statistics from group 1
5  Ybar1 <- mean(Y1)
6  s21   <- mean((Y1-Ybar1)^2)
7  n1    <- length(Y1)
8
9  # Statistics from group 2
10 Ybar2 <- mean(Y2)
11 s22   <- mean((Y2-Ybar2)^2)
12 n2    <- length(Y2)
13
14 # Posterior of the difference assuming equal variance
15 delta_hat <- Ybar2-Ybar1
16 s2        <- (n1*s21 + n2*s22)/(n1+n2)
17 scale     <- sqrt(s2)*sqrt(1/n1+1/n2)
18 df        <- n1+n2
19 cred_int  <- delta_hat + scale*qt(c(0.025,0.975),df=df)
20
21 # Posterior of delta assuming unequal variance using MC sampling
22 mu1       <- Ybar1 + sqrt(s21/n1)*rt(1000000,df=n1)
23 mu2       <- Ybar2 + sqrt(s22/n2)*rt(1000000,df=n2)
24 delta     <- mu2-mu1
25
26 hist(delta,main="Posterior distribution of the difference in
    means")
27 quantile(delta,c(0.025,0.975)) # 95% credible set

```

where $\hat{\sigma}^2 = (n_1 \hat{s}_1^2 + n_2 \hat{s}_2^2)/n$ is a pooled variance estimator. As with the one-sample model, the difference between the posterior for the known-variance versus unknown-variance cases is that an estimate of σ^2 is inserted in the posterior and the Gaussian distribution is replaced with Student's t distribution with n degrees of freedom. In the Bayesian analysis we did not “plug in” an estimate of σ^2 , rather, by accounting for its uncertainty and marginalizing over μ and σ^2 the posterior for δ happens to have a natural estimator of σ^2 in δ 's scale. Listing 4.1 implements this method.

Unequal variance: If the assumption that the variance is the same for both groups is violated, then the two-sample model can be extended as $Y_i \stackrel{iid}{\sim} \text{Normal}(\mu_1, \sigma_1^2)$ for $i = 1, \dots, n_1$ and $Y_i \stackrel{iid}{\sim} \text{Normal}(\mu_2, \sigma_2^2)$ for $i = n_1 + 1, \dots, n_1 + n_2$. Since no parameters are shared across the two groups, we can apply the one-sample model separately for each group to obtain

$$\mu_j | \mathbf{Y} \stackrel{indep}{\sim} t_{n_j}(\bar{Y}_j, s_j^2/n_j) \quad (4.8)$$

for $j = 1, 2$, and the posterior of the difference in means $\delta = \mu_2 - \mu_1$ follows. The posterior of δ is the difference between two Student t random variables, which does not in general have a simple form. However, the posterior can be approximated with arbitrary precision using Monte Carlo sampling as in Listing 4.1.

4.2 Linear regression

The multiple linear regression model with response (also called the dependent variable or outcome) Y_i and covariates (also called independent variables, predictors or inputs) $X_{i1}, \dots, X_{ip}$ is

$$Y_i = \sum_{j=1}^p X_{ij}\beta_j + \varepsilon_i, \quad (4.9)$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$ are the regression coefficients and the errors (also called the residuals) are $\varepsilon_i \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2)$. We assume throughout that $X_{i1} = 1$ for all i so that β_1 is the intercept (i.e., the mean if all other covariates are zero). This model includes as special cases the one-sample mean model in Section 4.1.1 (with $p = 1$ and $\beta_1 = \mu$) and the two-sample mean model in Section 4.1.2 (with $p = 2$, X_{i2} equal one if observation i is from the second group and zero otherwise, $\beta_1 = \mu$, and $\beta_2 = \delta$).

The coefficient β_j for $j > 0$ is the slope associated with the j^{th} covariate. For the remainder of this subsection we will assume that all $p - 1$ covariates (excluding the intercept term) have been standardized to have mean zero and variance one so the prior can be specified without considering the scales of the covariates. That is, if the original covariate j , $\tilde{X}_{ij}$, had sample mean $\bar{X}_j$ and standard deviation $\hat{s}_j$ then we set $X_{ij} = (\tilde{X}_{ij} - \bar{X}_j)/\hat{s}_j$. After standardization, the slope β_j is interpreted as the change in the mean response corresponding to an increase of one standard deviation unit ($\hat{s}_j$) in the original covariate. Similarly, $\beta_j/\hat{s}_j$ is the expected increase in the mean response associated with an increase of one in the original covariate. The model actually has $p + 1$ parameters (p regression coefficients and variance σ^2) so we temporarily use p as the number of regression parameters and not the total number of parameters in the model.

The likelihood function for the linear model with n observations is the product of n Gaussian PDFs with means $\sum_{j=1}^p X_{ij}\beta_j$ and variance σ^2 ,

$$f(\mathbf{Y}|\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma}} \exp \left[-\frac{1}{2\sigma^2} \left(Y_i - \sum_{j=1}^p X_{ij}\beta_j \right)^2 \right]. \quad (4.10)$$

Since maximizing the likelihood is equivalent to minimizing the negative log-likelihood, the maximum likelihood estimator for β can be written

$$\beta_{LS} = \arg \min_{\beta} \sum_{i=1}^n \left(Y_i - \sum_{j=1}^p X_{ij} \beta_j \right)^2. \quad (4.11)$$

Therefore, assuming Gaussian errors the maximum likelihood estimator is also the least squares estimator.

The model is conveniently written in matrix notation. Let $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ be the response vector of length n , and the design matrix $\mathbf{X}$ be the $n \times p$ matrix with the first column equal to the vector of ones for the intercept and column j having elements $X_{1j}, \dots, X_{nj}$. The linear regression model is then

$$\mathbf{Y} \sim \text{Normal}(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n) \quad (4.12)$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix with ones on the diagonal (so all responses have variance σ^2) and zeros off the diagonal (so the responses are uncorrelated). The usual least squares estimator in matrix notation is

$$\hat{\beta}_{LS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (4.13)$$

Note that the least squares estimator is unique only if $\mathbf{X}^T \mathbf{X}$ is full rank (i.e., $p < n$ and none of $\mathbf{X}$'s columns are redundant) and the estimator is poorly defined if $\mathbf{X}^T \mathbf{X}$ is not full rank. Assuming $\mathbf{X}$ is full rank, the sampling distribution used to construct frequentist confidence intervals and p-values is $\hat{\beta}_{LS} \sim \text{Normal}(\beta_0, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1})$, where β_0 is the true value.

4.2.1 Jeffreys prior

Conditional distribution with the variance fixed: Conditioned on σ^2 , the Jeffreys prior is $\pi(\beta) \propto 1$. This improper prior only leads to a proper posterior if $\mathbf{X}^T \mathbf{X}$ has full rank, which is the same condition needed for the least squares estimator to be unique. Assuming this condition is met, the posterior is

$$\beta | \mathbf{Y}, \sigma^2 \sim \text{Normal} \left[\hat{\beta}_{LS}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \right]. \quad (4.14)$$

The posterior mean is the least squares solution and the posterior covariance matrix is the covariance matrix of the sampling distribution of the least squares estimator. Therefore, the posterior credible intervals from this model will numerically match the confidence intervals from a least squares analysis with known error variance.

Unknown variance: With σ^2 unknown, the Jeffreys' prior is $\pi(\beta, \sigma^2) \propto (\sigma^2)^{-p/2-1}$ (Section 2.3). Assuming $\mathbf{X}^T \mathbf{X}$ has full rank, Appendix A.3 shows that

$$\beta | \mathbf{Y} \sim t_n \left[\hat{\beta}_{LS}, \hat{\sigma}^2 (\mathbf{X}^T \mathbf{X})^{-1} \right], \quad (4.15)$$

Listing 4.2

R code for Bayesian linear regression under the Jeffreys' prior.

```

1  # This code assumes:
2  # Y is the n-vector of observations
3  # X is the n x p matrix of covariates
4  # The first column of X is all ones for the intercept
5
6  # Compute posterior mean and 95% interval
7  beta_mean <- solve(t(X)%*%X)%*%t(X)%*%Y
8  sigma2    <- mean((Y-X%*%beta_mean)^2)
9  beta_cov  <- sigma2*solve(t(X)%*%X)
10 beta_scale <- sqrt(diag(beta_cov))
11 df        <- length(Y)
12 beta_025  <- beta_mean + beta_scale*qt(0.025,df=df)
13 beta_975  <- beta_mean + beta_scale*qt(0.975,df=df)
14
15 # Package the output
16 out       <- cbind(beta_mean,beta_025,beta_975)
17 rownames(out) <- colnames(X)
18 colnames(out) <- c("Mean","Q 0.025","Q 0.975")

```

where $\hat{\sigma}^2 = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{LS})^T(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{LS})/n$. That is, the marginal posterior of $\boldsymbol{\beta}$ follows the p -dimensional Student's t distribution with location $\hat{\boldsymbol{\beta}}_{LS}$, covariance matrix $\hat{\sigma}^2(\mathbf{X}^T\mathbf{X})^{-1}$, and n degrees of freedom.

A property of the multivariate t distribution is that the marginal distribution of each element is univariate t so that

$$\beta_j | \mathbf{Y} \sim t_n(\hat{\beta}_j, s_j^2) \quad (4.16)$$

where $\hat{\beta}_j$ is the j^{th} element of $\hat{\boldsymbol{\beta}}_{LS}$ and s_j^2 is the j^{th} diagonal element of $\hat{\sigma}^2(\mathbf{X}^T\mathbf{X})^{-1}$. Therefore, both the joint distribution of all the regression coefficients and marginal distribution of each regression coefficient belong to a known family of distributions, and can thus be computed without MCMC sampling. Listing 4.2 provides R code to compute the posterior mean and 95% intervals.

4.2.2 Gaussian prior

For high-dimensional cases with more predictors than observations the improper Jeffreys' prior does not lead to a valid posterior distribution and thus a proper prior is required. Even in less extreme cases with moderate p , a proper prior can stabilize the posterior and give better results than the improper prior. The conjugate prior (conditioned on σ^2) for $\boldsymbol{\beta}$ is $\boldsymbol{\beta} | \sigma^2 \sim \text{Normal}(\boldsymbol{\mu}, \sigma^2 \boldsymbol{\Omega})$. We include σ^2 in the prior variance to account for the scale of the response. Typically the prior is centered around zero,

Listing 4.3

JAGS code for multiple linear regression with Gaussian priors.

```

1  # Likelihood
2  for(i in 1:n){
3    Y[i] ~ dnorm(inprod(X[i,],beta[]),taue)
4  }
5  # Priors
6  beta[1] ~ dnorm(0,0.001) #X[i,1]=1 for the intercept
7  for(j in 2:p){
8    beta[j] ~ dnorm(0,taub*taue)
9  }
10 taue ~ dgamma(0.1, 0.1)
11 taub ~ dgamma(0.1, 0.1)

```

and so from here on we set $\boldsymbol{\mu} = \mathbf{0}$. This prior combined with the likelihood $\mathbf{Y} \sim \text{Normal}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$ gives posterior

$$\boldsymbol{\beta} | \mathbf{Y}, \sigma^2 \sim \text{Normal} \left[(\mathbf{X}^T \mathbf{X} + \boldsymbol{\Omega}^{-1})^{-1} \mathbf{X}^T \mathbf{Y}, \sigma^2 (\mathbf{X}^T \mathbf{X} + \boldsymbol{\Omega}^{-1})^{-1} \right], \quad (4.17)$$

which is a proper distribution as long as the prior is proper ($\boldsymbol{\Omega}$ is positive definite) even if $p > n$ or the covariates are perfectly collinear.

There are several choices for the prior covariance matrix $\boldsymbol{\Omega}$. The most common is the diagonal matrix $\boldsymbol{\Omega} = \tau^2 \mathbf{I}_p$, which is equivalent to the prior $\beta_j | \sigma^2 \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2 \tau^2)$. Under this prior the MAP estimator is the ridge regression estimator [44]

$$\boldsymbol{\beta}_R = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n \left(Y_i - \sum_{j=1}^p X_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p \beta_j^2, \quad (4.18)$$

where $\lambda = 1/\tau^2$. Ridge regression is often used to stabilize least squares problems when the number of predictors is large and/or the covariates are collinear. In ridge regression, the tuning parameter λ can be selected based on cross-validation. In a fully Bayesian analysis, τ^2 can either be fixed to a large value to give an uninformative prior, or given a conjugate inverse gamma prior as in Listing 4.3 to allow the data to determine how much to shrink the coefficients towards zero. If τ^2 is given a prior, then the intercept term β_1 should be given a different variance because it plays a different role than the other regression coefficients.

The form of the posterior simplifies under Zellner's g-prior [83] $\boldsymbol{\beta} | \sigma^2 \sim \text{Normal} \left[\mathbf{0}, \frac{\sigma^2}{g} (\mathbf{X}^T \mathbf{X})^{-1} \right]$ for $g > 0$. The conditional posterior is then

$$\boldsymbol{\beta} | \mathbf{Y}, \sigma^2 \sim \text{Normal} \left[c \hat{\boldsymbol{\beta}}_{LS}, c \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \right], \quad (4.19)$$

where $c = 1/(g + 1) \in (0, 1)$ is the shrinkage factor. This speeds computation

because $\hat{\beta}_{LS}$ and $(\mathbf{X}^T \mathbf{X})^{-1}$ can be computed once outside the MCMC sampler. The shrinkage factor c determines how strongly the posterior mean and covariance are shrunk towards zero. A common choice is $g = 1/n$ and thus $c = n/(n+1)$. Since Fisher's information matrix for the Gaussian distribution is the inverse covariance matrix, the prior contributes $1/n^{th}$ the information as the likelihood, and so this prior is called the unit information prior [49].

4.2.3 Continuous shrinkage priors

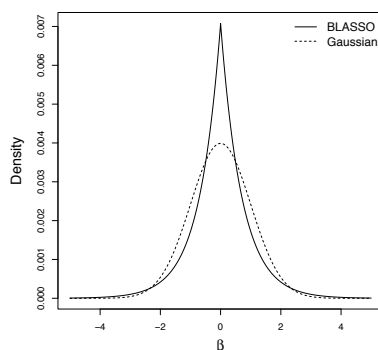
In regressions with many predictors it is often assumed that most of the p predictors have little effect on the response. The Gaussian prior can reflect this prior belief with a small prior variance, but a small prior variance has the negative side effect of shrinking the posterior mean of the important regression coefficients toward zero and thus inducing bias. An alternative is to use double exponential priors for the β_j that have (Figure 4.2) a peak at zero to reflect the prior belief that most predictors have no effect on the response, but a heavy tail to reflect the prior belief that there are a few predictors with strong effects. Listing 4.4 provides **JAGS** code to implement this model (although the **R** package **BLR** [21] is likely faster).

Assuming the model $Y_i \sim \text{Normal}(\sum_{j=1}^p X_{ij}\beta_j, \sigma^2)$ with double exponential priors $\beta_j \stackrel{iid}{\sim} \text{DE}(\lambda/\sigma^2)$ and thus prior PDF $\pi(\beta_j) \propto \exp(-\frac{\lambda}{2\sigma^2}|\beta_j|)$, then the maximum a posterior (MAP) estimate is (since maximizing the posterior is equivalent to minimizing twice the negative log posterior)

$$\hat{\beta}_{LASSO} = \arg \min_{\beta} \sum_{i=1}^n \left(Y_i - \sum_{j=1}^p X_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j|. \quad (4.20)$$

This is the famous LASSO [81] penalized regression estimator and thus the double exponential prior is often called the Bayesian LASSO prior. An attractive feature of this estimator is that some of the estimates may have $\hat{\beta}_j$ set exactly to zero, and this then performs variable selection simultaneously with estimation. In other words, the LASSO encodes the prior belief that some of the covariates are unimportant.

The double exponential prior is just one example of a shrinkage prior with peak at zero and heavy tails. The horseshoe prior [16] is $\beta_j | \lambda_j \sim \text{Normal}(0, \lambda_0^2 \lambda_j^2)$ where λ_0 is global variance common to all regression coefficients and λ_j is a local prior standard deviation specific to β_j . The local variances are given half-Cauchy prior (i.e., student-t prior with one degree of freedom restricted to be positive). This global-local prior is designed to shrink null coefficients towards zero by having small variance while the true signals have uninformative priors with large variance. The Dirichlet-Laplace prior [11] gives even more shrinkage towards zero by supplementing the Bayesian LASSO with local shrinkage parameters and a Dirichlet prior on the shrinkage parameters. The R2D2 prior [84] is another global-local shrinkage prior for

**FIGURE 4.2**

Comparison of the Gaussian and double exponential prior distributions. Below are the standard normal PDF and the double exponential PDF with parameters set to give mean zero and variance one.

the regression coefficients that is constructed so that the model's coefficient of determination (i.e., R-squared) has a beta prior. The R2D2 prior has the most mass at zero and heaviest tail among these priors. For an alternative approach that places positive prior probability on the regression coefficients being exactly zero see the spike-and-slab prior in Section 5.3.

4.2.4 Predictions

One use of linear regression is to make a prediction for a new set of covariates, $\mathbf{X}^{pred} = (X_1^{pred}, \dots, X_p^{pred})$. Given the model parameters, the distribution of

Listing 4.4

JAGS code for the Bayesian LASSO.

```

1  # Likelihood
2  for(i in 1:n){
3    Y[i] ~ dnorm(inprod(X[i,],beta[]),taue)
4  }
5  # Priors
6  beta[1] ~ dnorm(0,0.001)
7  for(j in 2:p){
8    beta[j] ~ ddexp(0,taub*taue)
9  }
10 taue ~ dgamma(0.1, 0.1)
11 taub ~ dgamma(0.1, 0.1)

```

Listing 4.5

JAGS code for linear regression predictions.

```

1  # Likelihood
2  for(i in 1:n){
3    Y[i] ~ dnorm(inprod(X[i,],beta[]),taue)
4  }
5  # Priors
6  beta[1] ~ dnorm(0,0.001)
7  for(j in 2:p){
8    beta[j] ~ dnorm(0,taub*taue)
9  }
10 taue ~ dgamma(0.1, 0.1)
11 taub ~ dgamma(0.1, 0.1)
12
13 # Predictions
14 for(i in 1:n_pred){
15   Y_pred[i] ~ dnorm(inprod(X_pred[i,],beta[]),taue)
16 }
17 # User must pass JAGS the covariates X_pred and integer n_pred
18 # JAGS returns PPD samples of Y_pred

```

the new response is $Y^{pred}|\beta, \sigma^2 \sim \text{Normal}(\sum_{j=1}^p X_j^{pred} \beta_j, \sigma^2)$. To properly account for parametric uncertainty, we should use the posterior predictive distribution (Section 1.5) that averages over the uncertainty in β and σ^2 . MCMC provides a means to sample from the PPD by making a sample from the predictive distribution for each of the $s = 1, \dots, S$ MCMC samples of the parameters, $Y^{(s)}|\beta^{(s)}, \sigma^{2(s)} \sim \text{Normal}(\sum_{j=1}^p X_j^{pred} \beta_j^{(s)}, \sigma^{2(s)})$, and using the S draws $Y^{(1)}, \dots, Y^{(S)}$ to approximate the PPD. The PPD is then summarized the same way as other posterior distributions, such as by the posterior mean and 95% interval. Similar approaches can use to analyze missing data as in Section 6.4.

Listing 4.5 gives JAGS code to make linear regression predictions. The matrix of predictors X_{pred} must be passed to JAGS and JAGS will return the predictions Y_{pred} . Making predictions with JAGS can slow the sampler and consume memory, and so it is often better to first perform MCMC sampling for the parameters using JAGS and then make predictions in R as in Listing 4.6.

4.2.5 Example: Factors that affect a home's microbiome

We use the data from [5], downloaded from <http://figshare.com/articles/1000homes/1270900>. The data are dust samples from the ledges above doorways from $n = 1,059$ homes (after removing samples with missing data; for missing data methods see Section 6.4) in the continental US. Bioinformatics processing detects the presence or absence of 763 species (technically

Listing 4.6

R code to use JAGS MCMC samples for linear regression predictions.

```

1  # INPUTS
2  #   beta_samples := S x p matrix of MCMC samples (from JAGS)
3  #   taue_samples := S x 1 matrix of MCMC samples (from JAGS)
4  #   X_pred       := n_pred x p matrix of prediction covariates
5
6  S      <- nrow(beta_samples)
7  n_pred <- nrow(X_pred)
8  Y_pred <- matrix(NA,S,n_pred)
9  sigma  <- 1/sqrt(taue_samples)
10
11 for(s in 1:S){
12   Y_pred[s,] ~ X_pred%*%beta_samples[s,]+rnorm(n_pred,0,sigma[s])
13 }
14
15 # OUTPUT
16 #   Y_pred       := S x n_pred matrix of PPD samples

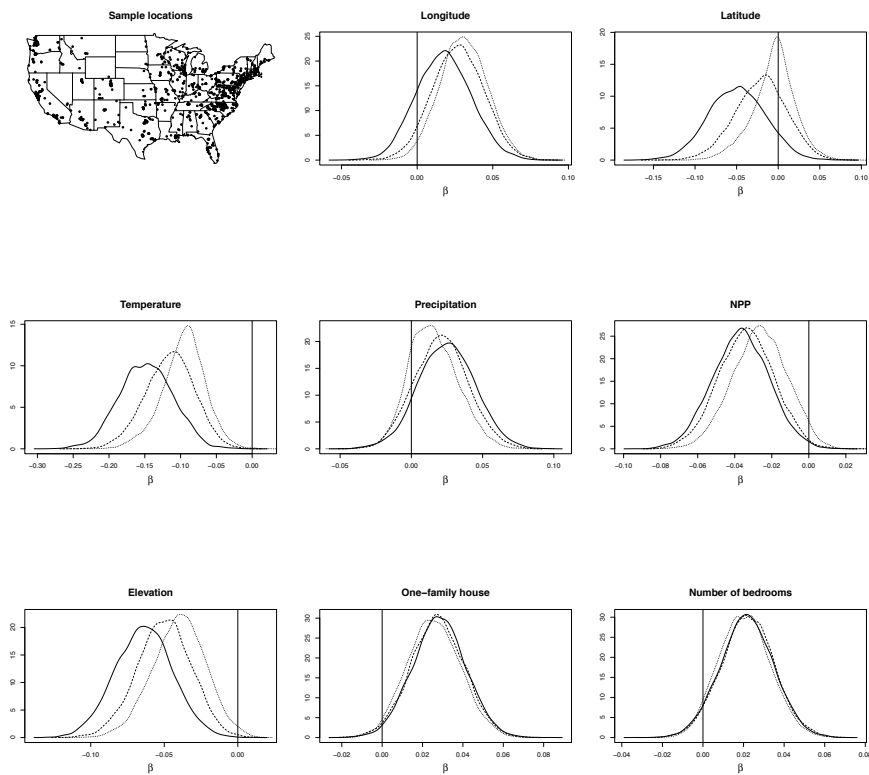
```

operational taxonomic units) of fungi. The response is the log of the number of fungi species present in the sample, which is a measure of species richness. The objective is to determine which factors influence a home's species richness. For each home, eight covariates are included in this example: longitude, latitude, annual mean temperature, annual mean precipitation, net primary productivity (NPP), elevation, the binary indicator that the house is a single-family home, and the number of bedrooms in the home. These covariates are all centered and scaled to have mean zero and variance one.

We apply the Gaussian model in Listing 4.3 first with $\beta_j \stackrel{iid}{\sim} \text{Normal}(0, 100^2)$ ("Flat prior") and then with $\beta_j | \sigma^2, \tau^2 \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2 \tau^2)$ with $\tau^2 \sim \text{InvGamma}(0.1, 0.1)$ ("Gaussian shrinkage prior"), and the Bayesian LASSO prior in Listing 4.4. For each of the three models we ran two MCMC chains with 10,000 samples in the burn-in and 20,000 samples after burn-in. Trace plots (not shown) showed excellent convergence and the effective sample size exceeded 1,000 for all parameters and all models.

The results are fairly similar for the three priors (Figure 4.3). In all three models, temperature, NPP, elevation, and single-family home are the most important predictors, with the most richness estimated to occur in single-family homes with low temperature, NPP, and elevation. In all three models, the sample with largest fitted value (i.e., the posterior mean of $\mathbf{X}\beta$) is a single-family home with three bedrooms in Montpelier, VT, and the sample with the smallest fitted value is a multiple-family home with two bedrooms in Tempe, AZ.

Although the results are not that sensitive to the prior in this analysis, there are some notable difference. For example, compared to the posterior under a flat priors, the posterior of the slope for latitude (top right panel in

**FIGURE 4.3**

Regression analysis of the richness of a home's microbiome. The first panel shows the sample locations and the remaining panels plot the posterior distributions of the regression coefficients, β_j . The three models are distinguished by their priors for β_j : the flat prior is $\beta_j \sim \text{Normal}(0, 100^2)$ (solid line), the Gaussian shrinkage prior is $\beta_j \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2 \tau^2)$ with $\tau^2 \sim \text{InvGamma}(0.1, 0.1)$ (dashed line), and the Bayesian LASSO is $\beta_j \stackrel{iid}{\sim} \text{DE}(0, \sigma^2 \tau^2)$ with $\tau^2 \sim \text{InvGamma}(0.1, 0.1)$ (dotted line).

Figure 4.3) is shrunk towards zero by the Gaussian shrinkage model, and the posterior density concentrates even more around the origin for the Bayesian LASSO prior. However, it is not clear from this plot which of these three fits is preferred; model comparison is discussed in Chapter 5.

4.3 Generalized linear models

Multiple linear regression assumes that the response variable is Gaussian and thus the mean response can be any real number. Many analyses do not conform to this assumption. For example, in Section 4.3.1 we analyze binary responses with support $\{0, 1\}$ and in Section 4.3.2 we analyze count data with support $\{0, 1, 2, \dots\}$. Clearly these data are not Gaussian and their mean cannot be any real number because the mean must be between zero and one for binary data and positive for count data. The generalized linear model (GLM) extends linear regression concepts to handle these non-Gaussian outcomes. There is a deep theory of GLMs (exponential families, canonical links, etc; see [54]), but here we focus only on casting the GLM in the Bayesian framework through a few examples.

The basic steps to selecting a GLM are (1) determine the support of the response and select an appropriate parametric family and (2) link the covariates to the parameters that define this family. As an example, consider the Gaussian linear regression model in Section 4.2. If the support of the response is $(-\infty, \infty)$, a natural parametric family is the Gaussian distribution. Of course, there are other families with this support and the fit of the Gaussian family to the data should be verified empirically. Once the Gaussian family has been selected, the covariates must be linked to one or both of the parameters, the mean or the variance. Let the linear combination of the covariates be

$$\eta_i = \sum_{j=1}^p X_{ij}\beta_j. \quad (4.21)$$

The linear predictor η_i can take any value in $(-\infty, \infty)$ depending on X_{ij} . Therefore, to link the covariates with the mean we can simply set $E(Y_i) = \eta_i$ as in standard linear regression. To complete the standard model we elect not to link the covariates with the variance, and simply set $V(Y_i) = \sigma^2$ for all i .

The function that links the linear predictor with a parameter is called the link function. Say that the parameter in the likelihood for the response is θ_i (e.g., $E(Y_i) = \theta_i$ or $V(Y_i) = \theta_i$), then the link function g is

$$g(\theta_i) = \eta_i. \quad (4.22)$$

The link function must be an invertible function that is well-defined for all permissible values of the parameter. For example, in the Gaussian case the

Listing 4.7

Model statements for several GLMs in JAGS.

```

1
2  # (a) Logistic regression
3    for(i in 1:n){
4      Y[i] ~ dbern(q[i])
5      logit(q[i]) <- inprod(X[i,],beta[])
6    }
7    for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
8
9  # (b) Probit regression
10   for(i in 1:n){
11     Y[i] ~ dbern(q[i])
12     probit(q[i]) <- inprod(X[i,],beta[])
13   }
14   for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
15
16  # (c) Poisson regression
17   for(i in 1:n){
18     Y[i] ~ dpois(lambda[i])
19     log(lambda[i]) <- inprod(X[i,],beta[])
20   }
21   for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
22
23  # (d) Negative binomial regression
24   for(i in 1:n){
25     Y[i] ~ dnegbin(q[i],m)
26     q[i] <- m/(m + lambda[i])
27     log(lambda[i]) <- inprod(X[i,],beta[])
28   }
29   for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
30   m ~ dgamma(0.1,0.1)
31
32  # (e) Zero-inflated Poisson
33   for(i in 1:n){
34     Y[i] ~ dpois(q[i])
35     q[i] <- Z[i]*lambda[i]
36     Z[i] ~ dbern(p[i])
37     log(lambda[i]) <- inprod(X[i,],beta[])
38     logit(p[i]) <- inprod(X[i,],alpha[])
39   }
40   for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
41   for(j in 1:p){alpha[j] ~ dnorm(0,0.01)}
42
43  # (f) Beta regression
44   for(i in 1:n){
45     Y[i] ~ dbeta(r*q[i],r*(1-q[i]))
46     logit(q[i]) <- inprod(X[i,],beta[])
47   }
48   for(j in 1:p){beta[j] ~ dnorm(0,0.01)}
49   r ~ dgamma(0.1,0.1)

```

link function for the mean is the identity function $g(x) = x$ so that the mean can be any real number. To link the covariates to the variance we must ensure that the variance is positive, and so natural-log function $g(x) = \log(x)$ is more appropriate. Link functions are not unique, and must be selected by the user. For example, the link function for the mean could be replaced by $g(x) = x^3$ and the link function for the variance could be replaced with $g(x) = \log_{10}(x)$.

Bayesian fitting of a GLM requires selecting the prior and computing the posterior distribution. The priors for the regression coefficients discussed for Gaussian data in Section 4.2 can be applied for GLMs. The posterior distributions for GLMs are usually too complicated to derive the posterior in closed-form and prove that, say, a particular prior distribution leads to a student-t posterior distribution with interpretable mean and covariance. However, much of the intuition developed with Gaussian linear models carries over to GLMs.

With non-Gaussian responses the full conditional distributions for the β_j are usually not conjugate and Metropolis sampling must be used. Maximum likelihood estimates can be used as initial values and the corresponding standard errors can suggest appropriate candidate distributions. In the examples in this chapter we use **JAGS** to carry out the MCMC sampling. **R** also has dedicated packages for Bayesian GLMs, such as the **MCMClogit** package for logistic regression (Section 4.3.1) that are likely faster than **JAGS**. With flat priors and a large sample it is even more efficient to evoke the Bayesian central limit theorem (Section 3.1.3) and approximate the posterior as Gaussian (e.g., using the **glm** function in **R**) and avoid MCMC altogether.

4.3.1 Binary data

Binary outcomes $Y_i \in \{0, 1\}$ occur frequently when the result of an experiment is recorded only as an indicator of a success or failure. For example, in Section 1.2 the response was the binary indicator that a test for HIV was positive. Binary variables must follow a Bernoulli distribution. The Bernoulli distribution has one parameter, the success probability $\text{Prob}(Y_i = 1) = q_i \in [0, 1]$. Since the parameter is a probability, the link function must have input range $[0, 1]$ and output range $[-\infty, \infty]$. Below we discuss two such link functions: logistic and probit.

Logistic regression: The logistic link function is

$$g(q) = \text{logit}(q) = \log\left(\frac{q}{1-q}\right). \quad (4.23)$$

This link function converts the event probability q first to the odds of the event, $q/(1-q) > 0$, and then to the log odds, which can be any real number. The logistic regression model is written

$$Y_i \stackrel{\text{indep}}{\sim} \text{Bernoulli}(q_i) \text{ and } \text{logit}(q_i) = \eta_i = \sum_{j=1}^J X_{ij}\beta_j. \quad (4.24)$$

The inverse logistic function is $g^{-1}(x) = \exp(x)/[1 + \exp(x)]$ and so the model can also be expressed as $Y_i \stackrel{\text{indep}}{\sim} \text{Bernoulli}(\exp(\eta_i)/[1 + \exp(\eta_i)])$. JAGS code for this model is in Listing 4.7a.

Because the log odds of the event that $Y_i = 1$ are linear in the covariates, β_j is interpreted as the increase in the log odds corresponding to an increase of one in X_j with all other covariates held fixed. Similarly, with all over covariates held fixed, increasing X_j by one multiplies the odds by $\exp(\beta_j)$. Therefore if $\beta_j = 2.3$, increasing X_j by one multiplies the odds by ten and if $\beta_j = -2.3$, increasing X_j by one divides the odds by ten. This interpretation is convenient for communicating the results and specifying priors. For example, if a change of one in the covariate is deemed a large change, then a standard normal prior may have sufficient spread to represent an uninformative prior.

Probit regression: There are many possible link functions from $[0, 1]$ to $[-\infty, \infty]$. In fact, the quantile function (inverse CDF) of any continuous random variable with support $[-\infty, \infty]$ would suffice. The link function in logistic regression is the quantile function of the logistic distribution. In probit regression, the link function is the Gaussian quantile function

$$Y_i \stackrel{\text{indep}}{\sim} \text{Bernoulli}(q_i) \text{ and } q_i = \Phi(\eta_i), \quad (4.25)$$

where Φ is the standard normal CDF (Listing 4.7b).

Unfortunately, the regression coefficients β_j in probit regression do not have a nice interpretation such as their log-odds interpretation in logistic regression. However, probit regression is useful because it leads to Gibbs sampling and can be used to model dependence between binary variables. Probit regression is equivalent to specifying latent variables $Z_i \stackrel{\text{indep}}{\sim} \text{Normal}(\eta_i, \sigma^2)$ and assuming that observing $Y_i = 1$ indicates that Z_i exceeds the threshold z . For example, Z_i might represent a patient's blood pressure and Y_i is the corresponding binary indicator that the patient has high blood pressure (defined as exceeding z). In this example, the probability that the patient has high blood pressure is

$$\begin{aligned} \text{Prob}(Y_i = 1) &= \text{Prob}(Z_i > z) \\ &= \text{Prob}[(Z_i - \eta_i)/\sigma > (z - \eta_i)/\sigma] \\ &= 1 - \Phi[(z - \eta_i)/\sigma] \\ &= \Phi[(\eta_i - z)/\sigma] \\ &= \Phi \left[\left(\sum_{j=1}^J X_{ij} \beta_j - z \right) / \sigma \right]. \end{aligned}$$

Since we never observe the latent variables Z_i we cannot estimate the threshold z or the variance σ^2 . The threshold z is coupled with the intercept because adding a constant to the threshold and subtracting the same constant from the intercept does not affect the event probabilities q_i . Therefore, the

threshold is typically set to $z = 0$. Similarly, multiplying σ and dividing the slopes β_j by the same constant does not affect the event probabilities, and so the variance is typically set to $\sigma^2 = 1$. This gives the usual probit regression model $q_i = \Phi(\eta_i)$.

In this formulation, the regression coefficients β have conjugate full conditional distributions conditioned on the latent Z_i . This leads to Gibbs sampling if the Z_i are imputed at each MCMC iteration [3]. Also, dependence between binary outcomes Y_i can be induced by a multivariate normal model for the latent variables Z_i .

4.3.2 Count data

Random variables with support $Y_i \in \{0, 1, 2, \dots\}$ often arise as the number of events that occur in a time interval or spatial region. For example, in Section 2.1 we analyze the number of NFL concussions by season. In this chapter, we will focus on modeling the mean as a function of the covariates. The link between the linear predictor and the mean must ensure that $E(Y_i) = \lambda_i \geq 0$. A natural link function is the log link,

$$\log(\lambda_i) = \sum_{j=1}^p X_{ij}\beta_j. \quad (4.26)$$

In this model, the mean is multiplied by $\exp(\beta_j)$ if X_j increases by one with all other covariates remaining fixed. Unlike binary data, specifying the mean does not completely determine the likelihood for count data. We discuss below two families of distributions for the likelihood function: Poisson and negative binomial.

Poisson regression: The Poisson regression model is

$$Y_i | \lambda_i \stackrel{\text{indep}}{\sim} \text{Poisson}(\lambda_i) \text{ where } \lambda_i = \exp \left(\sum_{j=1}^p X_{ij}\beta_j \right). \quad (4.27)$$

As with logistic regression, the regression coefficients $\beta_1, \dots, \beta_p$ can be given the priors discussed in Section 4.2, and Metropolis sampling can be used to explore the posterior (Listing 4.7c). A critical assumption of the Poisson model is that the mean and variance are equal, that is,

$$E(Y_i) = V(Y_i) = \lambda_i. \quad (4.28)$$

The distribution of a count is over-dispersed (under-dispersed) if its variance is greater than (less than) its mean. If over-dispersion is present, then the Poisson model is inappropriate and another model should be considered so that the posterior accurately reflects all sources of variability.

Negative binomial regression: One approach to accommodating over-dispersion is to incorporate gamma random variables $e_i \stackrel{iid}{\sim} \text{Gamma}(m, m)$

with mean 1 and variance $1/m$. Then

$$Y_i | e_i, \lambda_i, m \stackrel{\text{indep}}{\sim} \text{Poisson}(\lambda_i e_i). \quad (4.29)$$

Marginally over e_i , Y_i follows the negative binomial distribution,

$$Y_i | \lambda_i, m \sim \text{NegBinomial}(q_i, m) \quad (4.30)$$

with probability $q_i = m/(\lambda_i + m)$ and size m . The size m need not be an integer, but if it is and we envision a sequence of independent Bernoulli trials each with success probability q_i , then Y_i can be interpreted as the number of failures that occur before the m^{th} success. More importantly for handling over-dispersion is that $E(Y_i) = \lambda_i$ and $V(Y_i) = \lambda_i + \lambda_i^2/m > \lambda_i$. The parameter m controls over-dispersion. If m is large, then $e_i \approx 1$ and the model reduces to the Poisson regression model with $V(Y_i) \approx E(Y_i)$; if m is close to zero, then e_i has large variance and thus $V(Y_i) > E(Y_i)$. The over-dispersion parameter can be given a gamma prior, as in Listing 4.7d.

Zero-inflated Poisson: Another common deviation from the Poisson distribution is an excess of zeros. For example, say that Y_i is the number of fish caught by visitor i to a state park. It may be that most of the Y_i are zero because only a small proportion (say p) of the visitors fished, but the distribution of Y_i for those that did fish is Poisson with mean λ . The probability of an observation being zero is then $1 - p$ for the non-fishers plus p times the Poisson probability at zero for the fishers. The PMF corresponding to this scenario is

$$f(y|p, \lambda) = \begin{cases} (1 - p) + pf_P(0|\lambda) & \text{if } y = 0 \\ pf_P(y|\lambda) & \text{if } y > 0 \end{cases} \quad (4.31)$$

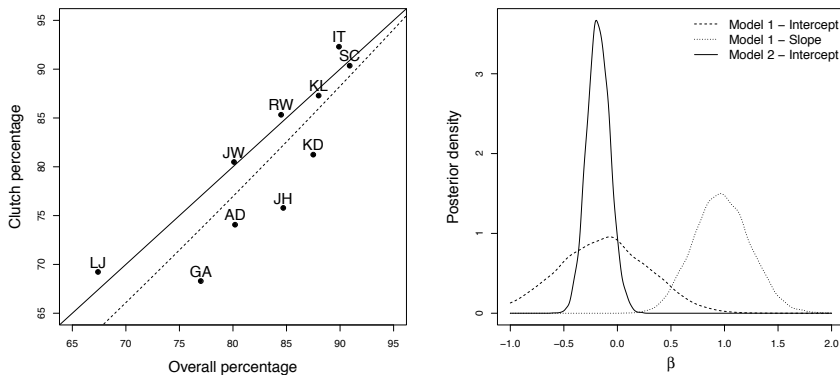
where f_P is the Poisson PMF. This is equivalent to the two-stage model

$$Y_i | Z_i, \lambda \sim \text{Poisson}(Z_i \lambda) \text{ and } Z_i \sim \text{Bernoulli}(p)$$

where Z_i is the latent indicator that visitor i fished and thus the mean of Y_i is zero for non-fishers with $Z_i = 0$. In this scenario, we do not observe the Z_i but this two-stage model gives the equivalent model to (4.31) but uses only standard distributions. As a result, the model can be coded in JAGS with covariates included in the mass at zero and the Poisson rate as in Listing 4.7e.

4.3.3 Example: Logistic regression for NBA clutch free throws

The table in Problem 17 of Section 1.6 gives the overall free-throw proportion ($q_i \in [0, 1]$), the number of made clutch shots (Y_i), and the number of attempted clutch shots (n_i) for players $i = 1, \dots, 10$. Figure 4.4 shows that most players have similar success rates in clutch and non-clutch situations, but some players appear to have less success in pressure situations. We fit logistic regression models to formally explore this relationship.

**FIGURE 4.4**

Logistic regression analysis of NBA free throws. The first panel shows the overall percentage of made free throws versus the percentage for clutch shots only for each player (denoted by the player's initials). The solid line is the $x = y$ lines and the dashed line is the fitted value from Model 2. The second plot is the posterior density for the slope and intercept from the Model 1 and the intercept from the model 2.

The support of the response is $Y_i \in \{0, 1, \dots, n_i\}$ and so we select a binomial likelihood $Y_i | p_i \sim \text{Binomial}(n_i, p_i)$ where p_i is the probability of player i making a clutch shot. The number of shots n_i is known, and so we link the covariate to the success probability p_i . The two models are

1. $\text{logit}(p_i) = \beta_1 + \beta_2 X_i$
2. $\text{logit}(p_i) = \beta_1 + X_i$

where $X_i = \text{logit}(q_i)$ is the log odds of making a regular free throw. If $\beta_1 = 0$ and $\beta_2 = 1$ in Model 1 or $\beta_1 = 0$ in Model 2, then the clutch performance equals the overall performance. To determine if this is the case, we fit the model using JAGS with two chains with burn-in 10,000 and 20,000 additional samples and with uninformative $\text{Normal}(0, 100)$ priors for all parameters. The model specification for Model 1 is

```
for(i in 1:10){
  Y[i] ~ dbinom(p[i], n[i])
  logit(p[i]) <- beta[1] + beta[2]*X[i]
}
beta[1] ~ dnorm(0, 0.01)
beta[2] ~ dnorm(0, 0.01)
```

The results are plotted in Figure 4.4. For Model 1, the slope is centered squarely on one and the intercept is centered slightly below zero, but both

TABLE 4.1

Beta regression of microbiome data. Posterior median and 95% intervals for the regression coefficients β_j and concentration parameter r .

	Median	95% Interval
Intercept	-1.01	(-1.05, -0.96)
Longitude	-0.21	(-0.28, -0.14)
Latitude	-0.08	(-0.22, 0.05)
Temperature	-0.15	(-0.30, -0.01)
Precipitation	0.03	(-0.04, 0.11)
NPP	-0.04	(-0.10, 0.02)
Elevation	-0.02	(-0.09, 0.05)
Single-family home	0.07	(0.02, 0.13)
Number of bedrooms	-0.08	(-0.13, -0.03)
r	7.97	(7.34, 8.66)

parameters have considerable uncertainty and so the model with $p_i = q_i$ remains plausible. However, the intercept in Model 2 is negative with posterior probability 0.96, so there is some evidence that even the best players in the NBA underperform in pressure situations. The fitted curve (dashed line) in Figure 4.4 (left panel) is $1/[1 + \exp(-\bar{\beta}_1 - X_i)]$, where $\bar{\beta}_1$ is the posterior mean, and shows that the clutch performance can be several percentage points lower than the overall percentage.

4.3.4 Example: Beta regression for microbiome data

In Section 4.2.5 we regressed the diversity of a sample's microbiome onto features of the home. Diversity was measured as the log of the number of the $L = 763$ species present in the sample. However, this measure fails to account for the relative abundance of the species. Let $A_{il} \geq 0$ be the abundance of species l in sample i . Another measure of the diversity is the proportion of the total abundance attributed to the most abundant species in sample i ,

$$Y_i = \frac{\max\{A_{i1}, \dots, A_{iL}\}}{\sum_{l=1}^L A_{il}} \in [0, 1]. \quad (4.32)$$

This measure is plotted in Figure 4.5 (top left).

Since Y_i is between 0 and 1, Gaussian linear regression is inappropriate. One option is to transform the response from support $[0, 1]$ to $(-\infty, \infty)$, e.g., $Y_i^* = \text{logit}(Y_i)$, and model the transformed data Y_i^* using a Gaussian linear model. Another option is to model the data directly using a non-Gaussian model. A natural model for a continuous variable with support $[0, 1]$ is the beta distribution. Since the mean must also be in $[0, 1]$, a logistic link can be used to relate the covariates to the mean response. Therefore, we fit the beta

regression model

$$Y_i|\boldsymbol{\beta}, r \sim \text{Beta}[rq_i, r(1 - q_i)] \text{ and } \text{logit}(q_i) = \mathbf{X}_i^T \boldsymbol{\beta}. \quad (4.33)$$

$Y_i|\boldsymbol{\beta}, r$ has mean q_i and variance $q_i(1 - q_i)/(r + 1)$, and so $r > 0$ determines the concentration of the beta distribution around the mean q_i .

Using priors $\beta_j \sim \text{Normal}(0, 100)$ and $r \sim \text{Gamma}(0.1, 0.1)$, we fit this model in JAGS using the code in Listing 4.7e (although sampling would likely be faster using the `betareg` package in R). Before fitting the model, the co-variates are standardized to have mean zero and variance one. Convergence was excellent for all parameters (the bottom left panel of Figure 4.5 shows the trace plot for r). The posterior distributions in Table 4.1 indicate that there is more diversity (smaller Y_i) on average in homes in cool regions in the east, and multiple-family homes with many bedrooms.

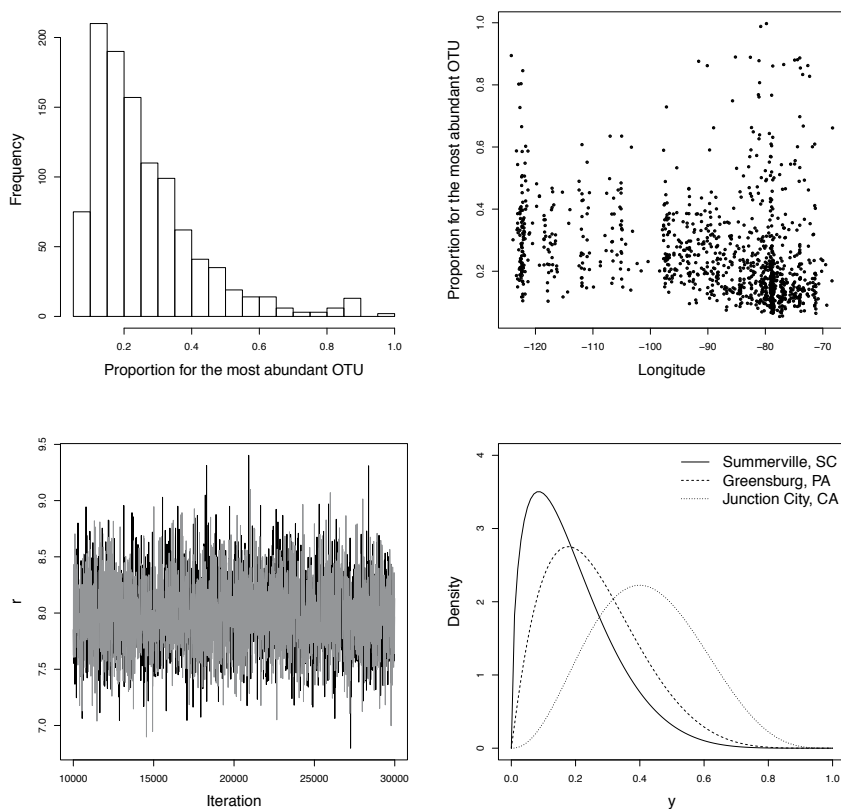
4.4 Random effects

The standard linear regression model assumes the same regression model applies to all observations. This assumption is tenuous if data are collected in groups. For example, Figure 4.6 plots the jaw bone density measurements of $n = 20$ children measured over the course of $m = 4$ visits. These 20 children represent only a random sample from a much larger population, but we can use these samples to make an inference about the larger population. If we let Y_{ij} be the j^{th} measurement for patient i , then (ignoring age as in the top right panel of Figure 4.6) the one-way random effects model is

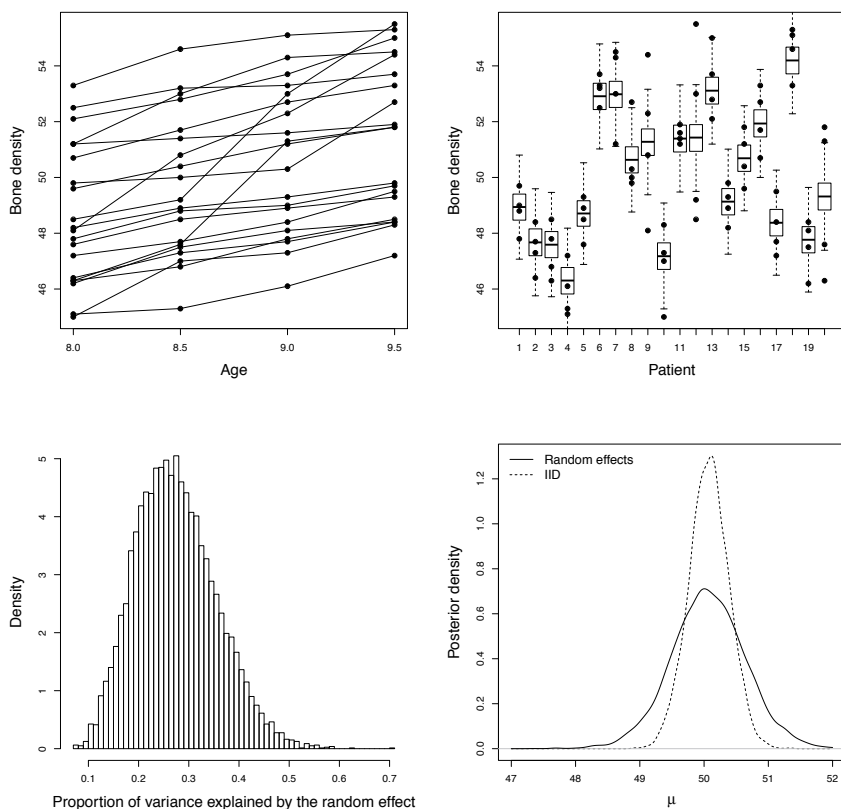
$$Y_{ij}|\alpha_i \stackrel{\text{indep}}{\sim} \text{Normal}(\alpha_i, \sigma^2) \text{ where } \alpha_i \stackrel{\text{iid}}{\sim} \text{Normal}(\mu, \tau^2). \quad (4.34)$$

The random effect α_i is the true mean for patient i , and the observations for patient i vary around α_i with variance σ^2 . The α_i are called random effects because if we repeated the experiment with a new sample of 20 children the α_i would change. In this model, the population of patient-specific means is assumed to follow a normal distribution with mean μ and variance τ^2 . The overall mean μ is a fixed effect because if we repeated the experiment with a new sample of 20 children from the same population it would not change. A linear model with both fixed and random effects is called a linear mixed model.

A Bayesian analysis of a random effects model requires priors for the population parameters. For example, Listing 4.8 provides JAGS code with conjugate priors $\mu \sim \text{Normal}(0, 100^2)$ and $\tau^2 \sim \text{InvGamma}(0.1, 0.1)$. The same algorithms (e.g., Gibbs sampling) can be used for random effects models as for the other models we have considered. In fact, computationally there is no need to distinguish between fixed and random effects (which can lead to

**FIGURE 4.5**

Beta regression for microbiome data. The top left panel shows the histogram of the observed proportions of abundance allocated to the most abundance OTU, and the top right panel plots this variable against the sample's longitude. The second row gives the trace plots (the two chains are different shades of gray) of the concentration parameter, r , and the fitted $\text{Beta}[\hat{r}\hat{q}, \hat{r}(1 - \hat{q})]$ density for three samples evaluated at the posterior mean for all parameters.

**FIGURE 4.6**

One-way random effect analysis of the jaw data. The dots in the top left panel show bone density at the four visits for each patient (connected by lines), the top right panel compares the observations (dots) and the posterior distribution of the subject random effect α_i (boxplot), the bottom left panel plots the posterior of the variance ratio $\tau^2/(\tau^2 + \sigma^2)$, and the final panel compares the posterior density of the mean μ from the random effects model versus independence model $Y_{ij} \stackrel{iid}{\sim} \text{Normal}(\mu, \sigma^2)$.

Listing 4.8

The one-way random effects model in JAGS.

```

1  # Likelihood
2  for(i in 1:n){for(j in 1:m){
3    Y[i,j] ~ dnorm(alpha[i],sig2_inv)
4  }}
5
6  # Random effects
7  for(i in 1:n){alpha[i] ~ dnorm(mu,tau2_inv)}
8
9  # Priors
10 mu ~ dnorm(0,0.0001)
11 sig2_inv ~ dgamma(0.1,0.1)
12 tau2_inv ~ dgamma(0.1,0.1)

```

confusion, [42]). Conceptually, however, fixed and random effects are distinct because fixed effects describe the population and a random effect describes an individual from the population. A common source of confusion is that fixed effects (such as μ in Listing 4.8) are treated as random variables in a Bayesian analysis. However, as with all parameters, the prior and posterior distributions for fixed effects reflect subjective uncertainty about the true values of these fixed but unknown parameters.

Unlike analyses such as linear regression where the main focus is on the mean and prediction of new observations, in random effects models the variance components (e.g., σ^2 and τ^2) are often the main focus of the analysis. Therefore it is important to scrutinize the priors used for these parameters. The inverse gamma prior is conjugate for the variance parameters which often leads to simple Gibbs updates. However, as shown in Figure 4.7, the inverse gamma prior with small shape parameter for a variance induces a prior for the standard deviation with prior PDF equal to zero at the origin. This excludes the possibility that there is no random effect variance, which may not be suitable in many problems.

As an alternative, [29] endorses a half-Cauchy (HC) prior for the standard deviation. The HC distribution is the student-t distribution with one degree of freedom restricted to be positive and has a flat PDF at the origin (Figure 4.7), which is usually a more accurate expression of prior belief. Listing 4.9 gives JAGS code for this prior. In this code the HC distribution is assigned directly to the standard deviations which breaks the conjugacy relationships for the variance components (this is easily handled by JAGS); [29] shows that conjugacy can be restored using a two-stage model. This code assumes the HC scale parameter is fixed at one. Because the Cauchy prior has a very heavy tail this gives 0.99 prior quantile equal to 63. Despite this wide prior range, the scale of the HC prior should be adjusted to the scale of the data.

Random effects induce correlation between observations from the same group. In the one-way random effects model, the covariance marginally over

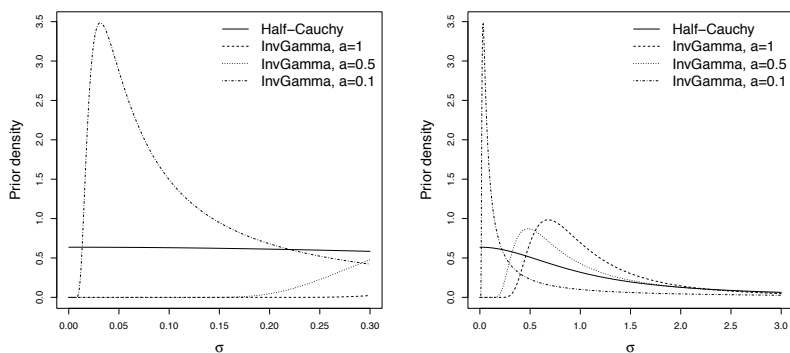
Listing 4.9

The one-way random effects model with half-Cauchy priors.

```

1  # Likelihood
2  for(i in 1:n){for(j in 1:m){
3    Y[i,j] ~ dnorm(alpha[i],sig2_inv)
4  }}
5
6  # Random effects
7  for(i in 1:n){alpha[i] ~ dnorm(mu,tau2_inv)}
8
9  # Priors
10 mu      ~ dnorm(0,0.0001)
11 sig2_inv <- pow(sigma1,-2)
12 tau2_inv <- pow(sigma2,-2)
13 sigma1   ~ dt(0, 1, 1)T(0,) # Half-Cauchy priors with
14 sigma2   ~ dt(0, 1, 1)T(0,) # location 0 and scale 1

```

**FIGURE 4.7**

Priors for a standard deviation. The half-Cauchy prior for σ and the prior induced for σ by inverse gamma priors on σ^2 with different shape parameters. All priors are scaled to have median equal 1. The two panels differ only by the range of σ being plotted.

the random effect α_i is

$$\text{Cov}(Y_{ij}, Y_{uv}) = \begin{cases} \tau^2 + \sigma^2 & i = u \text{ and } j = v \\ \tau^2 & i = u \text{ and } j \neq v \\ 0 & i \neq u. \end{cases} \quad (4.35)$$

The variance of each observation is $\tau^2 + \sigma^2$ and includes both the variance in the random effect (τ^2) and variability around the mean (σ^2). Since two observations from the same patient share a common random effect, they have covariance τ^2 and thus correlation $\tau^2/(\tau^2 + \sigma^2)$. Observations from different patients have no common source of variability and are thus uncorrelated.

Returning to the bone density data, Figure 4.6 plots the posterior approximated with two chains each with 30,000 samples and the first 10,000 discarded as burn-in assuming the model in Listing 4.8. The posterior distribution of the proportion of the variance attributed to the random effect $\tau^2/(\tau^2 + \sigma^2)$ ranges from 0.1 to 0.5 (bottom left panel). Accounting for this correlation between observations from the same patient affects the posterior of the fixed effect, μ (bottom right). The posterior variance of μ is larger for the random effects model compared to the posterior of μ for the independence model $Y_{ij} \stackrel{iid}{\sim} \text{Normal}(\mu, \sigma^2)$. This is expected because there is less information about the mean in 4 repeated measurements for 20 patients than 80 measurements from 80 different patients.

To test for prior sensitivity, we refit the model using a half-Cauchy prior in Listing 4.9. In this case the prior for the variance components had little effect on the results. The 95% posterior credible sets σ and τ are (1.24, 1.78) and (1.75, 3.51) respectively using inverse gamma priors compared to (1.24, 1.77) and (1.72, 3.45) for the half-Cauchy priors.

Random slopes model: The one-way random effect model that ignores age is naive because bone density clearly increases with age (top left panel of Figure 4.6). Adding a time trend to the model accounts for this,

$$Y_{ij} | \alpha_i \stackrel{indep}{\sim} \text{Normal}(\alpha_i + X_j \beta, \sigma^2) \text{ where } \alpha_i \stackrel{iid}{\sim} \text{Normal}(\mu, \tau^2), \quad (4.36)$$

X_j is the child's age at visit j , and β is the fixed age trend.

This model assumes that each child has a different intercept but the same slope. However, Figure 4.6 indicates that the rate of increase over time varies by patient. Therefore we could model the patient-specific slopes as random effects

$$Y_{ij} | \alpha_i \stackrel{indep}{\sim} \text{Normal}(\alpha_{i1} + \alpha_{i2} X_j, \sigma^2) \text{ where } \alpha_i \stackrel{iid}{\sim} \text{Normal}(\beta, \Omega), \quad (4.37)$$

and the random intercept and slope for patient i is $\alpha_i = (\alpha_{i1}, \alpha_{i2})^T$. The mean vector β includes the population mean intercept and slope, and is thus a fixed effect. The 2×2 population covariance matrix Ω determines the variation of the random effects over the population. To complete the Bayesian model we specify prior $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$, $\beta \sim \text{Normal}(\mathbf{0}, 100^2 \mathbf{I}_2)$, and $\Omega \sim$

Listing 4.10

Random slopes model in JAGS.

```

1  # Likelihood
2  for(i in 1:n){for(j in 1:m){
3    Y[i,j] ~ dnorm(alpha[i,1]+alpha[i,2]*age[j],tau)
4  }}
5
6  # Random effects
7  for(i in 1:n){
8    alpha[i,1:2] ~ dmnorm(beta[1:2],Omega_inv[1:2,1:2])
9  }
10
11 # Priors
12 tau ~ dgamma(0.1,0.1)
13 for(j in 1:2){beta[j] ~ dnorm(0,0.0001)}
14 Omega_inv[1:2,1:2] ~ dwish(R[,],2.1)
15
16 R[1,1]<-1/2.1
17 R[1,2]<-0
18 R[2,1]<-0
19 R[2,2]<-1/2.1

```

InvWishart(2.1, $\mathbf{I}_2/2.1$). The inverse Wishart prior for the covariance matrix has prior mean $\mathbf{I}_2$, the 2×2 identity matrix (see Appendix A.1). JAGS code for this random-slopes model is in Listing 4.10.

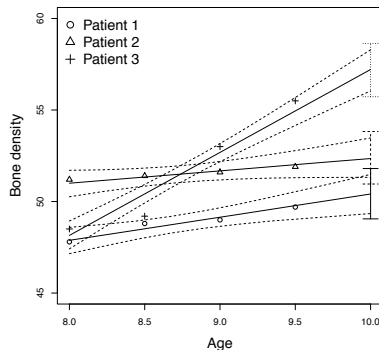
The posterior mean of the population covariance matrix is

$$E(\boldsymbol{\Omega}|\mathbf{Y}) = \begin{bmatrix} 91.78 & -10.14 \\ -10.14 & 1.23 \end{bmatrix} \quad (4.38)$$

and the posterior 95% interval for the correlation $\text{Cor}(\alpha_{i1}, \alpha_{i2}) = \Omega_{12}/\sqrt{\Omega_{11}\Omega_{22}}$ is (-0.98, -0.89). Therefore there is a strong negative dependence between the intercept and slope, indicating that bone density increases rapidly for children with low bone density at age 8, and vice versa.

Figure 4.8 plots the posterior distribution of the fitted values $\alpha_{i1} + \alpha_{i2}X$ for X between 8 and 10 years for three patients. For each patient and each age, we compute the 95% interval using the quantiles of the S posterior samples $\alpha_{i1}^{(s)} + \alpha_{i2}^{(s)}X$. We also plot the posterior predictive distribution (PPD) for the measured bone density at age 10. The PPD is approximated by sampling $Y_i^{*(s)} \sim \text{Normal}(\alpha_{i1}^{(s)} + \alpha_{i2}^{(s)}10, \sigma^{2(s)})$ at each iteration and then computing the quantiles of the S predictions. The PPD accounts for both uncertainty in the patient's random effect $\boldsymbol{\alpha}_i$ and measurement error with variance σ^2 . The intervals in Figure 4.8 suggest that uncertainty in the random effects is the dominant source of variation.

Marginal models: Inducing correlation by conditioning on random effects is equivalent to a marginal model (Section 4.5.4) that does not include random

**FIGURE 4.8**

Mixed effects analysis of the jaw data. The observed bone density (points) for three subjects versus the posterior median (solid lines) and 95% intervals (dashed lines) of the fitted value $\alpha_{i1} + \alpha_{i2}X$ for X ranging from 8–10 years, and 95% credible intervals (vertical lines at Age=10) of the posterior predictive distributions for the measured response at age $X = 10$.

effects but directly specifies correlation between observation in the same group. For example, the one-way random effects model

$$Y_{ij}|\alpha_i \stackrel{\text{indep}}{\sim} \text{Normal}(\alpha_i, \sigma^2) \text{ where } \alpha_i \stackrel{\text{iid}}{\sim} \text{Normal}(\mu, \tau^2) \quad (4.39)$$

is equivalent to the marginal model

$$\mathbf{Y}_i \sim \text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad (4.40)$$

where $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{im})^T$ is the data vector for group i , $\boldsymbol{\mu} = (\mu, \dots, \mu)^T$ is the mean vector, and $\boldsymbol{\Sigma}$ is the covariance matrix with $\tau^2 + \sigma^2$ on the diagonals and τ^2 elsewhere. An advantage of the marginal approach is that we no longer have to estimate the random effects α_i ; a disadvantage is that for large data sets and complex correlation structure the mean vector and especially the covariance matrix can be large which slows computation.

In the hierarchical representation, the elements of $\mathbf{Y}_i$ are independent and identically distributed given α_i ; in the marginal model the m observations from group i are no longer independent, but they remain exchangeable, i.e., their distribution is invariant to permuting their order. The concept of exchangeability plays a fundamental role in constructing hierarchical models. The representation theorem by Bruno de Finetti states that any infinite sequence of exchangeable variables can be written as independent and identically distributed conditioned on some latent distribution. Therefore, this important type of dependent data can be modeled using a simpler hierarchical model.

4.5 Flexible linear models

As discussed in Section 4.2, the multiple linear regression model of response Y_i onto covariates $X_{i1}, \dots, X_{ip}$ is

$$Y_i = \sum_{j=1}^p X_{ij}\beta_j + \varepsilon_i, \quad (4.41)$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$ are the regression coefficients and the errors are $\varepsilon_i \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2)$. This model makes four key assumptions:

- (1) Linearity: The mean of $Y_i | \mathbf{X}_i$ is linear in $\mathbf{X}_i$
- (2) Equal variance: The residual variance (σ^2) is the same for all i
- (3) Normality: The errors ε_i are Gaussian
- (4) Independence: The errors ε_i are independent

In real analyses, most if not all of these assumptions will be violated to some extent. Minor violations will not invalidate statistical inference, but glaring model misspecifications should be addressed. This chapter provides Bayesian remedies to model misspecification (each subsection addresses one of the four assumptions above). A strength of the Bayesian paradigm is that these models can be fit by simply adding a few lines of **JAGS** code and do not require fundamentally new theory or algorithms.

4.5.1 Nonparametric regression

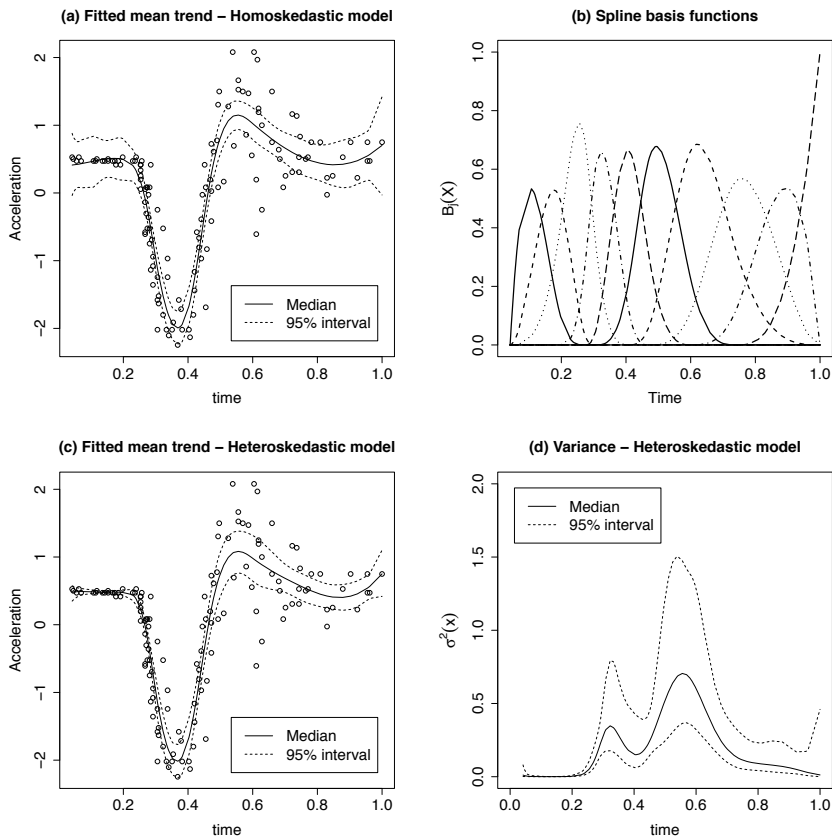
The linearity assumption can be relaxed by using a general expression of the regression of Y_i onto $\mathbf{X}_i$,

$$Y_i = g(\mathbf{X}_i) + \varepsilon_i = g(X_{i1}, \dots, X_{ip}) + \varepsilon_i \quad (4.42)$$

where g is the mean function and $\varepsilon_i \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2)$. Parametric regression specifies the mean function g as a parametric function of a finite number of parameters, e.g., in linear regression $g(\mathbf{X}_i) = \mathbf{X}_i\boldsymbol{\beta}$. A linear mean function is often a sufficient and interpretable first-order approximation, but more complex relationships between variables can be fit using a more flexible model.

For example, consider the data from the **mcycle** R package plotted in Figure 4.9. The predictor, X_i , is the time since a motorcycle makes impact (scaled to the unit interval) and the response, Y_i , is the acceleration of a monitor on the head of a crash test dummy. Clearly a linear model will not fit these data well: the mean is flat for the first quarter of the experiment, then dramatically dips, rebounds, and then plateaus until the end of the experiment.

A fully nonparametric model allows for any continuous function g . A model

**FIGURE 4.9**

Nonparametric regression for the motorcycle data. Panel (a) plots the time since impact (scaled to be between 0 and 1) and the acceleration (g) along with the posterior median and 95% interval for the mean function from the homoskedastic fit; Panel (b) shows the $J = 10$ spline basis functions $B_j(X)$; Panels (c) and (d) show the posterior median and 95% intervals for the mean and variance functions from the heteroskedastic model.

this flexible requires infinitely many parameters. We will focus on semiparametric models that specify the mean function in terms of a finite number of parameters in a way that increasing the number of parameters can approximate any function g . For example, if there is only $p = 1$ covariate (X) we could fit a J^{th} order polynomial function

$$g(X) = \sum_{j=0}^J X^j \beta_j. \quad (4.43)$$

This model has $J + 1$ parameters and by increasing J the polynomial function can approximate any continuous g .

There are many Bayesian semiparametric/nonparametric regression models, including Gaussian process regression [67], Bayesian adaptive regression trees [18], neural networks [61] and regression splines [19]. The simplest approach is arguably regression splines. In spline regression we construct nonlinear functions of the original covariates, and use these constructed covariates as the predictors in multiple linear regression. Denote the J constructed covariates as $B_1(\mathbf{X}), \dots, B_J(\mathbf{X})$. In polynomial regression $B_j(X) = X^j$, but there are many other choices. For example, Figure 4.9b plots $J = 10$ B-spline basis functions. These functions are appealing because they are smooth and local, i.e., non-zero only for some values of X . The model is simply the multiple linear regression model (Section 4.2) with the $B_1(X), \dots, B_J(X)$ as the covariates,

$$Y_i \sim \text{Normal}[g(X_i), \sigma^2] \text{ and } g(X_i) = \beta_0 + \sum_{j=1}^J B_j(X_i) \beta_j.$$

Note that each basis function in Figure 4.9b has $B_j(0) = 0$, and so an intercept (β_0) is required. By increasing J , any smooth mean function can be approximated as a linear combination of the B-spline basis functions.

Motorcycle example: To fit the mean curve to the data plotted in Figure 4.9a, we use $J = 10$ B-spline basis functions and priors $\beta_j \sim \text{Normal}(0, \tau^2 \sigma^2)$ and $\sigma^2, \tau^2 \sim \text{InvGamma}(0.1, 0.1)$. The model is fit using MCMC with the code in Listing 4.3. In this code, the basis functions have been computed in using the `bs` package in R and passed to JAGS as $x_{ij} = B_j(X_i)$. For each iteration, we compute $g(X_i)$ for all $i = 1, \dots, n$ as a function of that iteration's posterior sample of β . This produces the entire posterior distribution of the mean function g for all n sample points (and any other X we desire), and Figure 4.9a plots the posterior median and 95% interval of g at each sample point. The fitted model accurately captures the main trend including the valley around $X = 0.4$.

We selected $J = 10$ basis functions because this degree of model complexity visually seemed to fit the data well. Choosing smaller J would give a smoother estimate of g and choosing larger J would give a rougher estimate. Clearly a more rigorous approach to selecting the number of basis functions is needed, and this is discussed in Chapter 5.

Listing 4.11

Model statement for heteroskedastic Gaussian regression.

```

1  for(i in 1:n){
2    Y[i] ~ dnorm(mu[i],prec[i])
3    mu[i]  <- inprod(x[i,],beta[])
4    prec[i] <- 1/sig2[i]
5    sig2[i] <- exp(inprod(x[i,],alpha[]))
6  }
7  for(j in 1:p){beta[j] ~ dnorm(0,taub)}
8  for(j in 1:p){alpha[j] ~ dnorm(0,taua)}
9  taub ~ dgamma(0.1,0.1)
10 taua ~ dgamma(0.1,0.1)

```

4.5.2 Heteroskedastic models

The standard linear regression analysis assumes a homoskedastic variance $V(\varepsilon_i) = \sigma^2$ for all i . A more flexible heteroskedastic model allows the covariates to affect both the mean and the variance. A natural approach is to build a linear model for the variance as a function of the covariates so that $V(\varepsilon_i) = \sigma^2(\mathbf{X}_i)$. Since the variance is positive, we must transform the linear predictor to be positive before linking to the variance. For example,

$$\log[\sigma^2(\mathbf{X}_i)] = \sum_{j=1}^p X_{ij}\alpha_j, \quad (4.44)$$

or equivalently $\sigma^2(\mathbf{X}_i) = \exp(\sum_{j=1}^p X_{ij}\alpha_j)$. The parameter α_j determines the effect of the covariate j on the variance and must be estimated. **JAGS** code for this model is in Listing 4.11.

Motorcycle example: The variance of the observations about the mean trend in Figure 4.9a clearly depends on X , with small variance at the beginning of the experiment and large variance in the middle. To capture this heteroskedasticity in a flexible way, we model the log variance using the same J B-spline basis functions used for the mean,

$$g(X) = \beta_0 + \sum_{j=1}^p B_j(X)\beta_j \text{ and } \log[\sigma^2(X)] = \alpha_0 + \sum_{j=1}^p B_j(X)\alpha_j, \quad (4.45)$$

where $\beta_j \sim \text{Normal}(0, \sigma_b^2)$ and $\alpha_j \sim \text{Normal}(0, \sigma_a^2)$. The hyperparameters have uninformative priors $\sigma_a^2, \sigma_b^2 \sim \text{InvGamma}(0.1, 0.1)$.

The pointwise 95% intervals of $\sigma^2(X)$ in Figure 4.9d suggest that the variance is indeed small at the beginning of the experiment and increases with X . Comparing the posterior distributions of the mean trend for the homoskedastic (Figure 4.9a) and heteroskedastic (Figure 4.9c) models, the posterior means are similar but the heteroskedastic model produces more realistic 95% intervals with widths that vary according the pattern of the error variance. Therefore, it

appears that properly quantifying uncertainty about the mean trend requires a realistic model for the error variance.

4.5.3 Non-Gaussian error models

Most Bayesian regression models assume Gaussian errors, but more flexible methods are easily constructed. For example, to accommodate heavy tails the errors could be modelled using a student-t or double-exponential (Laplace) distributions. To further allow for asymmetry, generalizations such as the skew-t and asymmetric Laplace distributions would be used. Listing 4.12a provides code for regression with student-t errors.

In most analysis, a suitable parametric distribution can be found. However, this process is subjective and difficult to automate. Just as a mixture of conjugate priors (Section 2.1.8) can be used to approximate virtually any prior distribution, the mixture of normals distribution can be used to approximate virtually any residual distribution. The mixture of normals density for ε is

$$f(\varepsilon) = \sum_{k=1}^K \pi_k \phi(\varepsilon; \theta_k, \tau_1^2), \quad (4.46)$$

where K is the number of mixture components, $\pi_k \in (0, 1)$ is the probability on mixture component k , and $\phi(\varepsilon; \theta, \tau^2)$ is the Gaussian PDF with mean θ and variance τ^2 . This model is equivalent to the clustering model where

$$Y_i | g_i \sim \text{Normal} \left(\sum_{j=1}^p X_{ij} \beta_j + \theta_{g_i}, \sigma^2 \right) \quad (4.47)$$

and $g_i \in \{1, \dots, K\}$ is the cluster label for observation i with $\text{Prob}(g_i = k) = \pi_k$. By letting the number of mixture components increase to infinity, any distribution can be approximated, and by selecting priors $\theta_k \stackrel{iid}{\sim} \text{Normal}(0, \tau_2^2)$, $\tau_1^2, \tau_2^2 \sim \text{InvGamma}$, and $(\pi_1, \dots, \pi_K) \sim \text{Dirichlet}$ all full conditional distributions for all parameters are conjugate permitting Gibbs samples (Listing 4.12b).

For a fixed number of mixture components (K) the mixture-of-normals model is a semiparametric estimator of the density $f(\varepsilon)$. There is a rich literature on nonparametric Bayesian density estimation [37]. The most common model is the Dirichlet process mixture model that has infinitely many mixture components and a particular model for the mixture probabilities.

4.5.4 Linear models with correlated data

For data with a natural ordering such as spatial or temporal data modeling the correlation between observations is important to obtain valid inference and make accurate predictions. Generally, the Bayesian linear model with

Listing 4.12

Model statement for Gaussian regression with non-normal errors.

```

1
2  # (a) Regression with student-t errors
3  for(i in 1:n){
4    Y[i] ~ dt(mu[i],tau,df)
5    mu[i] <- inprod(X[i,],beta[])
6  }
7  for(j in 1:p){beta[j] ~ dnorm(0,taub)}
8  tau ~ dgamma(0.1,0.1)
9  df ~ dgamma(0.1,0.1)
10
11
12 # (b) Regression with mixture-of-normals errors
13 for(i in 1:n){
14   Y[i] ~ dnorm(mu[i]+theta[g[i]],tau1)
15   mu[i] <- inprod(X[i,],beta[])
16   g[i] ~ dcat(pi[])
17 }
18 for(k in 1:K){theta[k] ~ dnorm(0,tau2)}
19 for(j in 1:p){beta[j] ~ dnorm(0,tau3)}
20 tau1 ~ dgamma(0.1,0.1)
21 tau2 ~ dgamma(0.1,0.1)
22 tau3 ~ dgamma(0.1,0.1)
23 pi[1:K] ~ ddirch(alpha[1:K])

```

correlated errors is

$$\mathbf{Y} \sim \text{Normal}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}). \quad (4.48)$$

A Bayesian analysis of correlated data hinges on correctly specifying the correlation structure to capture say spatial or temporal correlation. Given the correlation structure and priors for the correlation parameters, standard Bayesian computational tools can be used to summarize the posterior. Correlation parameters usually will not have conjugate priors and so Metropolis–Hastings sampling is used. An advantage of the Bayesian approach for correlated data is that using MCMC sampling we can account for uncertainty in the correlation parameters for prediction or inference on other parameters, whereas maximum likelihood analysis often uses plug-in estimates of the correlation parameters and thus underestimates uncertainty.

Gun control example: The data for this analysis come from Kalesan et. al. (2016) [47]. The response variable, Y_i , is the log firearm-related death rate per 10,000 people in 2010 in state i (excluding Alaska and Hawaii). This is regressed onto five potential confounders: log 2009 firearm death rate per 10,000 people; firearm ownership rate quartile; unemployment rate quartile; non-firearm homicide rate quartile; and firearm export rate quartile. The covariate of interest is the number of gun control laws in effect in the state. This gives $p = 6$ covariates.

We first fit the usual Bayesian linear regression model

$$Y_i = \beta_0 + \sum_{j=1}^p X_{ij}\beta_j + \varepsilon_i \quad (4.49)$$

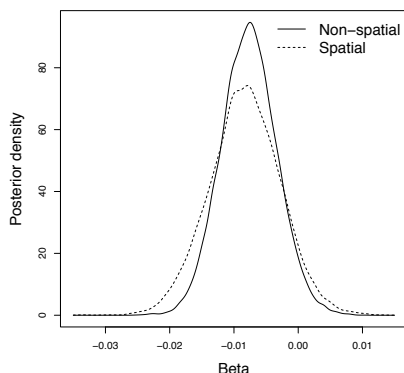
with independent errors $\varepsilon_i \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2)$ and uninformative priors. The posterior density of the regression coefficient corresponding to the number of gun laws is plotted in Figure 4.10. The posterior probability that the coefficient is negative is 0.96, suggesting a negative relationship between the number of gun laws and the firearm-related death rate.

The assumption of independent residuals is questionable because neighboring states may be correlated. Spatial correlation may stem from guns being brought across state borders or from missing covariates (e.g., attitudes about and use of guns) that vary spatially. Research has shown that accounting for residual dependence can have a dramatic effect on regression coefficient estimates [43].

We decompose the residual covariance $\text{Cov}[(\varepsilon_1, \dots, \varepsilon_n)^T] = \boldsymbol{\Sigma}$ as

$$\boldsymbol{\Sigma} = \tau^2 \mathbf{S} + \sigma^2 \mathbf{I}_n, \quad (4.50)$$

where $\tau^2 \mathbf{S}$ is the spatial covariance and $\sigma^2 \mathbf{I}_n$ is the non-spatial covariance. There are many spatial correlation models (e.g., [4]) that allow the correlation between two states to decay with the distance between the states. For example, a common model is to assume the correlation between states decays

**FIGURE 4.10**

Effect of gun-control legislation on firearm-related death rate. Posterior distribution of the coefficient associated with the number of gun-control laws in a state from the spatial and non-spatial model of the states' firearm-related death rate.

exponentially with the distance between them. However, quantifying the distance between irregularly shaped states is challenging, and so we model spatial dependence using adjacencies. Let $A_{ij} = 1$ if states i and j share a border and $A_{ij} = 0$ if $i = j$ or the states are not neighbors. The spatial covariance follows the conditionally autoregressive model $S = (M - \rho A)^{-1}$, where A is the adjacency matrix with (i, j) element A_{ij} and M is the diagonal matrix with the i^{th} diagonal element equal to the number of states that neighbor state i . The parameter $\rho \in (0, 1)$ is not the correlation between adjacent sites, but determines the strength of spatial dependence with $\rho = 0$ corresponding to independence.

The posterior mean (standard deviation) of the spatial dependence parameter ρ is 0.38 (0.25), and so the residual spatial dependence in these data is not strong. However, the posterior of the regression coefficient of interest in Figure 4.10 is noticeably wider for the spatial model than the non-spatial model. The posterior probability that the coefficient is negative lowers from 0.96 from the non-spatial model to 0.93 for the spatial model. Therefore, while accounting for residual dependence did not qualitatively change the results, this example illustrates that the chosen model for the residuals can affect the posterior of the regression coefficients.

Jaw bone density example: A possible correlation structure for the longitudinal data in Figure 4.6 (top left) is to assume that correlation decays with the time between visits. A first-order autoregression correlation structure is $\text{Cor}(Y_{ij}, Y_{ik}) = \rho^{|j-k|}$. Denoting the vector of m observations for patient

Listing 4.13

Random slopes model with autoregressive dependence in JAGS.

```

1  # Likelihood
2  for(i in 1:n){
3    Y[i,1:m] ~ dnmnorm(mn[i,1:m],SigmaInv)
4    for(j in 1:m){mn[i,j] <- alpha[i,1]+alpha[i,2]*age[j]}
5  }
6  SigmaInv[1:m,1:m] <- inverse(Sigma[1:m,1:m])
7  for(j in 1:m){for(k in 1:m){
8    Sigma[j,k] <- pow(rho,abs(k-j))/tau
9  }}
10
11 # Random effects
12 for(i in 1:n){alpha[i,1:2] ~ dnmnorm(beta[1:2],Omega[1:2,1:2])}
13
14 # Priors
15 tau ~ dgamma(0.1,0.1)
16 for(j in 1:2){beta[j] ~ dnorm(0,0.0001)}
17 rho ~ dunif(0,1)
18 Omega[1:2,1:2] ~ dwish(R[,],2.1)
19
20 R[1,1]<-1/2.1
21 R[1,2]<-0
22 R[2,1]<-0
23 R[2,2]<-1/2.1

```

i as $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{im})^T$, the $m \times m$ covariance matrix Σ has (j, k) element equal to $\sigma^2 \rho^{|j-k|}$. The random slope model for the mean in matrix notation is $E(\mathbf{Y}_i | \alpha_i) = \mathbf{X} \alpha_i$, where $\mathbf{X}$ is the $m \times 2$ matrix with the first column equal to the vector of ones for the intercept and the second column equal to the ages $X_1, \dots, X_m$ for the slope. The likelihood is then

$$\mathbf{Y}_i | \alpha_i \stackrel{\text{indep}}{\sim} \text{Normal}(\mathbf{X} \alpha_i, \Sigma) \quad (4.51)$$

with random effects distribution $\alpha_i \stackrel{iid}{\sim} \text{Normal}(\beta, \Omega)$. The correlation parameter is given prior $\rho \sim \text{Uniform}(0, 1)$ and all other priors are the same as other fits. JAGS code is given in Listing 4.13.

The posterior median of the correlation parameter ρ is 0.85 and the posterior 95% interval is (0.46, 0.96) so there is evidence of correlation that cannot be explained by the patient-specific linear trend. Including autoregressive correlation has only a modest effect on the posterior distribution of the fixed effects: the 95% posterior intervals for the random effect model with independent errors are (29.9, 38.3) for β_1 and (1.33, 2.38) for β_2 compared to (30.0, 37.3) for β_1 and (1.45, 2.31) for β_2 for the autoregressive model.

This model with random intercept, random slope and autoregressive correlation structure is now very complex. There are almost as many parameters

as observations and multiple explanations of dependence (random effects and residual correlation). Perhaps in this case all of these terms are necessary and can be estimated from this relatively small data set, but a simpler yet adequate model is preferred for computational purposes and because simpler models are easier to explain and defend. Model comparisons and tests of model adequacy are the topics of Chapter 5.

4.6 Exercises

1. A clinical trial gave six subjects a placebo and six subjects a new weight loss medication. The response variable is the change in weight (pounds) from baseline (so -2.0 means the subject lost 2 pounds). The data for the 12 subjects are:

Placebo	Treatment
2.0	-3.5
-3.1	-1.6
-1.0	-4.6
0.2	-0.9
0.3	-5.1
0.4	0.1

Conduct a Bayesian analysis to compare the means of these two groups. Would you say the treatment is effective? Is your conclusion sensitive to the prior?

2. Load the classic Boston Housing Data in R:

```
> library(MASS)
> data(Boston)
> ?Boston
```

The response variable is `medv`, the median value of owner-occupied homes (in \$1,000s), and the other 13 variables are covariates that describe the neighborhood.

- (a) Fit a Bayesian linear regression model with uninformative Gaussian priors for the regression coefficients. Verify the MCMC sampler has converged, and summarize the posterior distribution of all regression coefficients.
- (b) Perform a classic least squares analysis (e.g., using the `lm` function in R). Compare the results numerically and conceptually with the Bayesian results.

- (c) Refit the Bayesian model with double exponential priors for the regression coefficients, and discuss how the results differ from the analysis with uninformative priors.
 - (d) Fit a Bayesian linear regression model in (a) using only the first 500 observations and compute the posterior predictive distribution for the final 6 observations. Plot the posterior predictive distribution versus the actual value for these 6 observations and comment on whether the predictions are reasonable.
3. Download the 2016 Presidential Election data from the book's website. Perform Bayesian linear regression with the response variable for county i being the difference between the percentage of the vote for the Republican candidate in 2016 minus 2012 and all variables in the object $\mathbf{X}$ as covariates.
- (a) Fit a Bayesian linear regression model with uninformative Gaussian priors for the regression coefficients and summarize the posterior distribution of all regression coefficients.
 - (b) Compute the residuals $R_i = Y_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}$ where $\hat{\boldsymbol{\beta}}$ is the posterior mean of the regression coefficients. Are the residuals Gaussian? Which counties have the largest and smallest residuals, and what might this say about these counties?
 - (c) Include a random effect for the state, that is, for a county in state $l = 1, \dots, 50$,

$$Y_i | \alpha_l \sim \text{Normal}(\mathbf{X}_i \boldsymbol{\beta} + \alpha_l, \sigma^2)$$

where $\alpha_l \stackrel{iid}{\sim} \text{Normal}(0, \tau^2)$ and τ^2 has an uninformative prior. Why might adding random effects be necessary? How does adding random effects affect the posterior of the regression coefficients? Which states have the highest and lowest posterior mean random effect, and what might this imply about these states?

4. Download the US gun control data from the book's website. These data are taken from the cross-sectional study in [47]. For state i , let Y_i be the number of homicides and N_i be the population.
- (a) Fit the model $Y_i | \boldsymbol{\beta} \sim \text{Poisson}(N_i \lambda_i)$ where $\log(\lambda_i) = \mathbf{X}_i \boldsymbol{\beta}$. Use uninformative priors and $p = 7$ covariates in $\mathbf{X}_i$: the intercept, the five confounders $\mathbf{Z}_i$, and the total number of gun laws in state i . Provide justification that the MCMC sampler has converged and sufficiently explored the posterior distribution and summarize the posterior of $\boldsymbol{\beta}$.
 - (b) Fit a negative binomial regression model and compare with the results from Poisson regression.

- (c) For the Poisson model in (a), compute the posterior predictive distribution for each state with the number of gun laws set to zero. Repeat this with the number of gun laws set to 25 (the maximum number). According to these calculations, how would the number of deaths nationwide be affected by these policy changes? Do you trust these projections?
5. Download the titanic dataset from R,

```
library("titanic")
dat <- titanic_train
?titanic_train
```

Let $Y_i = 1$ if passenger i survived and $Y_i = 0$ otherwise. Perform a Bayesian logistic regression of the survival probability onto the passenger's age, gender (dummy variable) and class (two dummy variables). Summarize the effect of each covariate.

6. The T. rex growth chart data plotted in Figure 3.7 has $n = 6$ observations with weights (kg) 29.9, 1761, 1807, 2984, 3230, 5040, and 5654 and corresponding ages (years) 2, 15, 14, 16, 18, 22, and 28. Since weight must be positive, the gamma family of distributions is a reasonable model for these data. Describe a model with gamma likelihood and log mean that increases linearly with age. Approximate the posterior using MCMC, summarize the posterior distribution of all model parameters, and plot the data versus the fitted mean curve.
7. Consider the one-way random effects model $Y_{ij}|\alpha_i, \sigma^2 \sim \text{Normal}(\alpha_i, \sigma^2)$ and $\alpha_i \sim \text{Normal}(0, \tau^2)$ for $i = 1, \dots, n$ and $j = 1, \dots, m$. Assuming conjugate priors $\sigma^2, \tau^2 \sim \text{InvGamma}(a, b)$, derive the full conditional distributions of α_1 , σ^2 , and τ^2 and outline (but do not code) an MCMC algorithm to sample from the posterior.
8. Load the Gambia data in R:

```
> library(geoR)
> data(gambia)
> ?gambia
```

The response variable Y_i is the binary indicator that child i tested positive for malaria (**pos**) and the remaining seven variables are covariates.

- (a) Fit the logistic regression model

$$\text{logit}[\text{Prob}(Y_i = 1)] = \sum_{j=1}^p X_{ij}\beta_j$$

with uninformative priors for the β_j . Verify that the MCMC sampler has converged and summarize the effects of the covariates.

- (b) In this dataset, the 2,035 children reside in $L = 65$ unique locations (defined by the x and y coordinates in the dataset). Let $s_i \in \{1, \dots, L\}$ be the label of the location for observation i . Fit the random effects logistic regression model

$$\text{logit}[\text{Prob}(Y_i = 1)] = \sum_{j=1}^p X_{ij}\beta_j + \alpha_{s_i} \text{ where } \alpha_l \stackrel{iid}{\sim} \text{Normal}(0, \tau^2)$$

and the β_j and τ^2 have uninformative priors. Verify that the MCMC sampler has converged; explain why random effects might be needed here; discuss and explain any differences in the posteriors of the regression coefficients that occur when random effects are added to the model; plot the posterior means of the α_l by their spatial locations and suggest how this map might be useful to malaria researchers.

9. Download the `babynames` data in R and compute the log odds of a baby being named “Sophia” each year after 1950:

```
library(babynames)
dat <- babynames
dat <- dat[dat$name=="Sophia" &
          dat$sex=="F" &
          dat$year>1950,]
yr <- dat$year
p <- dat$prop
t <- dat$year - 1950
Y <- log(p/(1-p))
```

Let Y_t denote the sample log-odds in year $t + 1950$. Fit the following time series (auto-regressive order 1) model to these data:

$$Y_t = \mu_t + \rho(Y_{t-1} - \mu_{t-1}) + \varepsilon_t$$

where $\mu_t = \alpha + \beta t$ and $\varepsilon_t \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2)$. The priors are $\alpha, \beta \sim \text{Normal}(0, 100^2)$, $\rho \sim \text{Uniform}(-1, 1)$, and $\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$.

- Give an interpretation of each of the four model parameters: α , β , ρ , and σ^2 .
- Fit the model using JAGS for $t > 1$, verify convergence, and report the posterior mean and 95% interval for each parameter.
- Plot the posterior predictive distribution for Y_t in the year 2020.

10. Open and plot the galaxies data in R using the code below,

```
> library(MASS)
> data(galaxies)
> ?galaxies
> Y <- galaxies
> hist(Y,breaks=25)
```

Model the observations $Y_1, \dots, Y_{82}$ using a mixture of $K = 3$ normal distributions. For each the S MCMC iterations evaluate the density function on the grid $y \in \{5000, 5100, \dots, 40000\}$ (351 points in the grid), giving an $S \times 351$ matrix of posterior samples. Plot the posterior median and 95% credible set of the density function at each of the 351 grid values. Does this mixture model fit the data well?